Mandatory Disclosure and Housing Market Dynamics

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Abstract

Standard unraveling theory predicts that mandatory disclosure separates high-quality from low-quality products: silence signals bad news and non-disclosers should trade at-or-below the worst disclosed category. We document the opposite in the French housing market. Following the 2021 enforcement of mandatory Energy Performance Certificate (EPC) disclosure, listings without an EPC score trade at a premium of approximately 7 percentage points (apartments) and 3 percentage points (houses) *above* disclosed low-efficiency (EFG-rated) properties. We develop a model in which unraveling fails because non-disclosure pools two economically distinct populations: informed sellers strategically concealing unfavorable information, and uninformed sellers who genuinely do not know their property's energy performance. When buyers cannot distinguish between these groups, silence becomes ambiguous rather than damning, and low-quality sellers benefit from pooling with the genuinely uninformed. We show that the market's ability to price quality information depends on the composition of both sides: the share of informed sellers (π) and the share of quality-sensitive buyers (ρ). Full unraveling requires both to be high; for intermediate values, non-disclosure premia persist in equilibrium. We identify the model using a unique combination of residential listing records, transaction data, and the ADEME registry of certified energy audits, a third-party database that lets us directly observe each seller's information status at listing time, the latent variable that buyers cannot see. Our framework generates a closed-form minimum incentive-compatible subsidy for energy renovations that varies with local market composition. A central policy result is that the renovation threshold's apparent dependence on the hiding cost is illusory in the empirically dominant partial-pooling regime: the equilibrium indifference identity causes c_0 to cancel, and the threshold collapses to the verified-quality premium. Renovation subsidies, hiding-cost enforcement, and verification enforcement (audit-infrastructure investment, π) act on three distinct margins of the seller's problem and are complements rather than substitutes, contrary to a textbook perfect-substitutability intuition. Uniform national subsidies are generically inefficient.

Keywords: Energy Performance Certificates, Mandatory Disclosure, Housing Markets, Unraveling, Information Asymmetry
JEL Code: Q48, R31, Q52, D83, R38, H23

1 Introduction

Standard unraveling theory offers a stark prediction: in markets with verifiable quality information, sellers with good products disclose, silence signals bad news, and disclosure unravels until it is effectively universal (Grossman, 1981; Milgrom, 1981). The classical implication is that any seller who fails to disclose should be priced at-or-below the worst disclosed category, since buyers are entitled to expect the worst. The French housing market presents a sharp challenge to this view. Since 2021, France has required sellers to disclose Energy Performance Certificate (EPC) scores in all property listings. Yet four years after implementation, 48.8% of apartment listings lack a certified energy audit on record, and properties listed without an EPC score trade at a difference-in-differences premium of roughly 7 log points for apartments and 3 log points for houses *above* disclosed low-efficiency (EFG-rated) properties (Figure 11). Of these non-disclosing listings, 3.5% are strategic hiders: sellers who hold a certified audit but withhold it. The remainder are genuinely uninformed. The market does not punish opacity: it rewards it.

The puzzle is sharpened by the broader empirical record on energy labels in housing markets. A large body of work establishes that energy-efficient properties command meaningful price premia wherever EPCs have been studied: Brounen and Kok (2011) document a 2 to 5% premium for A/B-rated homes in the Netherlands following that country's 2008 mandate; Hyland et al. (2013) find premia of up to 9% in Ireland; Fuerst et al. (2016) confirm similar gradients in Wales; Frondel et al. (2018) show that Germany's shift from voluntary to mandatory disclosure reduced prices for non-disclosed listings, precisely as classical theory predicts; and Myers et al. (2022) demonstrate

that mandatory disclosure requirements in the United States increase the capitalization of energy efficiency and stimulate renovation investment. The consensus reading of this literature is that EPC mandates make quality salient, rewarding efficiency and penalizing waste. What none of these papers can explain is why *silence* (the deliberate omission of an EPC score) commands a premium. If anything, it should be the worst signal of all.

The gap lies in a foundational assumption that the classical framework and the EPC literature share: non-disclosure is always a *choice*. A seller who has not yet obtained a certified energy audit cannot disclose, regardless of her property's true performance. Dye (1985) first identified this distinction in the context of corporate earnings announcements, showing that when firms may simply lack private information, the market cannot interpret silence as unambiguously bad news. Jung and Kwon (1988) formalize the resulting equilibria: partial pooling (in which strategically hiding low-quality sellers shelter behind genuinely uninformed non-disclosers) is generically stable. When buyers observe a missing EPC score, they face irreducible ambiguity: they cannot determine whether the seller is concealing unfavorable information or simply lacks a certified audit. Silence becomes *noise*, not a signal, and its informativeness depends entirely on the composition of the non-disclosing pool.

Yet Dye-type models are set in corporate finance and accounting, where investors are treated as homogeneous in how much they value the disclosed signal. Housing markets are fundamentally different. A buyer purchasing a vacation property in the south of France, occupied a few weeks per year, may rationally ignore EPC scores; a buyer of a primary residence in northern France with cold winters will price energy costs directly. Loewenstein et al. (2014) survey evidence that disclosure interventions systematically fail when recipients are inattentive or heterogeneous in their responses; Ben-Shahar and Schneider (2011) document this pattern across many regulated markets, from mortgage disclosures to nutritional labels; and Fishman and Hagerty (2003) show formally that mandatory disclosure can be ineffective, or even welfare-reducing, when buyers differ in their capacity to process the disclosed information. When a large share of buyers is effectively indifferent to the EPC signal, the revenue penalty from concealment is diluted, sustaining hiding even when buyers hold pessimistic beliefs about the non-disclosing pool.

We formalize this intuition in a model with two key departures from the classical framework. First, we allow for **seller heterogeneity in information**: some sellers know their property's energy performance, while others do not. Second, we allow for **buyer heterogeneity in preferences**: some buyers care about energy efficiency, while others do not. These two dimensions, parameterized by π (the share of informed sellers) and ρ (the share of quality-sensitive buyers), jointly determine whether markets unravel, partially separate, or sustain pooling equilibria with premia to opacity. When both π and ρ are high, the classical benchmark is recovered. For intermediate values (our empirically relevant regime), partial separation with a persistent non-disclosure premium is the generic outcome.

A distinctive feature of this paper is the vantage point from which we observe the disclosure game. Testing a theory of strategic concealment requires seeing both sides of the information asymmetry simultaneously: the seller's true information, the seller's strategic choice, and the market's response. Most empirical work in this space sees only one or two of these. Studies of information asymmetry in housing infer agent advantages from equilibrium price deviations (Levitt and Syverson, 2008); analyses of certification intermediaries characterise incentives without observing individual information endowments (Lizzeri, 1999); and work on the energy efficiency gap must proxy for seller awareness through indirect variation in energy prices (Myers, 2020). We observe all three dimensions directly. Our dataset combines (i) **residential listing records** covering hundreds of thousands of French properties, with listing prices, characteristics, and each seller's disclosure decision; (ii) **transaction records** capturing actual sale prices and time-on-market, allowing us to trace the market's response from listing through closing; and (iii) the **ADEME registry of certified energy audits**, an official third-party database maintained by France's environment and energy management agency, compiled independently of the listing market. By matching properties across all three sources, we directly observe the latent variable K (whether a seller possesses a certified audit at listing time) that buyers can never see. This lets us classify every listing into one of four empirically meaningful groups: truthful disclosers, strategic hiders (informed sellers with a certified audit who omit it), genuinely uninformed sellers (no audit exists), and the rare misreporters (audit exists but displayed score diverges). The result is a dataset in which the full information structure of the disclosure game is observed, not inferred: a vantage point that, to our knowledge, is unique in the empirical disclosure literature.

We then extend the model to a policy context. When energy performance is endogenous (sellers can under-

take costly renovation), the renovation decision pits renovate-and-disclose against hiding: disclose-low is weakly dominated by hide because of the non-disclosure premium, so the renovation threshold $\bar{R} = \Pi(d{=}1, \theta{=}1) - \Pi(d{=}0; e^*) + c_0$ comes from a renovate-versus-hide comparison. The $+c_0$ term in this formula appears to give renovation subsidies and hiding-cost enforcement a textbook perfect-substitutability structure. We show that this appearance is misleading. In the empirically dominant partial-pooling regime, $\Pi(d{=}0; e^*)$ is itself an equilibrium revenue that adjusts to c_0 through the equilibrium hiding rate, and the indifference identity that defines the regime forces $\Pi(d{=}0; e^*) - c_0 = \Pi(d{=}1, \theta{=}0)$. Substituting, the renovation threshold reduces to the *verified-quality premium* $\Pi(d{=}1, \theta{=}1) - \Pi(d{=}1, \theta{=}0)$ and is invariant to c_0. The information-based wedge between private and social incentives is a new mechanism for the energy efficiency gap (Allcott and Greenstone, 2012; Gerarden et al., 2017), but its policy implication departs from textbook substitutability: the regulator confronts *three* distinct instruments acting on three different margins: renovation subsidies S, hiding-cost enforcement (raising c_0), and verification enforcement (raising π through audit-infrastructure investment). Subsidies shift the renovation propensity of informed low-quality sellers; hiding-cost enforcement shifts the composition of their disclosure decisions but does not move the renovation threshold in partial pooling; verification enforcement expands the share of the housing stock that is informed and therefore subject to the renovation decision in the first place. The three instruments are complements that scale each other's reach, not substitutes. This sharpens the policy logic underlying the documented free-riding in France's renovation tax credits (Nauleau, 2014): efficient targeting requires conditioning jointly on the verified-quality premium and the local verification penetration π, not on hiding-cost enforcement intensity (Gillingham and Palmer, 2014).

The remainder of the paper is organized as follows. Section 2 documents four stylized facts that any candidate theory of EPC disclosure in French housing markets must rationalize, drawing on our matched listings–ADEME panel. Section 3 develops the theoretical framework, characterizing equilibrium disclosure strategies as a function of (π, ρ, c_0) and establishing the conditions under which non-disclosure premia persist. Section 5 describes the data construction and documents the direct observation of each seller's information status via the ADEME registry match. Section 6 presents the difference-in-differences identification strategy around the July 2021 reform and discusses the exclusion restriction. Section 7 reports the structural estimates of ρ and c_0 at the market level, together with robustness checks and reduced-form validation. Section 8 derives the closed-form optimal renovation subsidy and the enforcement-subsidy equivalence result, and evaluates the efficiency cost of France's current uniform national policy.

2 Setting and Stylized Facts

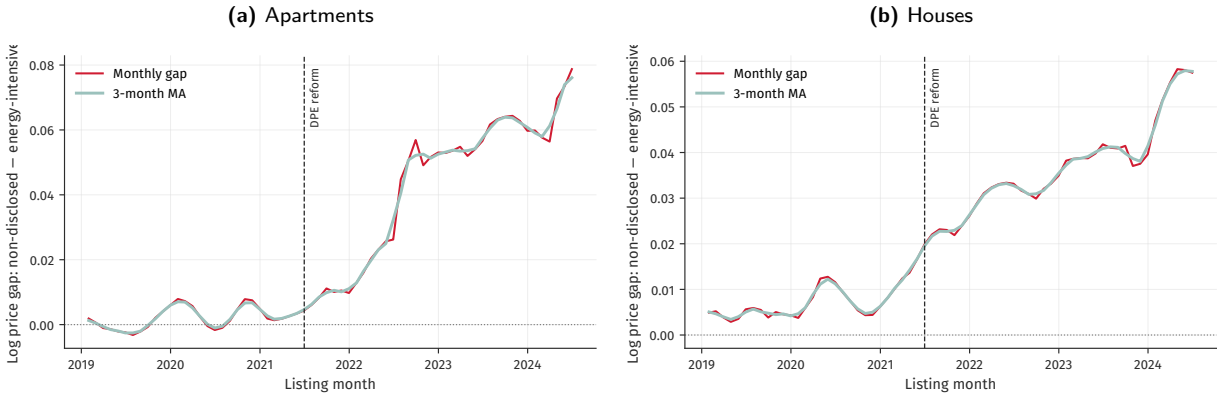
This section presents four stylized facts that motivate the theoretical framework developed in Section 3. The facts are drawn from a panel of 843,621 apartment listings and 687,878 house listings covering metropolitan France between July 2021 and December 2024, sourced from SeLoger, France's leading residential listings platform, and matched at the property level to the registry maintained by the *Agence de la Transition Écologique* (ADEME), France's national agency for ecological transition. The ADEME registry is the official, legally authoritative repository of every certified DPE issued in France; because audit certification is centralized there rather than left to the listing platform or the seller, the registry provides an independent, third-party measure of whether each property has been audited at the time of listing. Matching listings against this registry lets us observe directly, for the first time in the empirical disclosure literature, whether each seller is informed about her property's energy performance, separately from whether she chooses to disclose. This constitutes the central data contribution of the paper. Details of data construction, matching procedure, and sample restrictions are deferred to Section 5; here we focus on the empirical regularities themselves. Each figure presents apartments and houses side by side. The two segments differ markedly in the level of audit coverage and in the magnitude of the documented effects, but they display the same qualitative patterns, a robustness that we exploit throughout the paper. Each fact is paired with a specific structural feature that the model in Section 3 must accommodate.

2.1 Fact 1: A Non-Disclosure Premium Emerged with the 2021 Reform

The starting puzzle is that silence about energy performance commands a price premium relative to the worst certified scores in both property types. Figure 1 plots the volume-weighted log price gap between non-disclosing listings and certified energy-intensive (E/F/G) listings, by month, between January 2019 and June 2024, separately for apartments (panel (a)) and houses (panel (b)). Before the reform, the gap fluctuates near zero in both segments (a mean of $+0.002$ log points for apartments and $+0.008$ for houses), consistent with buyers treating absent EPCs

as roughly equivalent to the worst displayed scores when the DPE carried no legal weight. After July 2021, when DPE certificates became legally opposable under *loi Climat et Résilience*, the gap widens monotonically in both panels and stabilizes around $+0.044$ log points for apartments and $+0.037$ log points for houses by mid-2024. The reform thus produces a sharp, persistent, and economically large opacity premium in both segments. Classical unraveling predicts the opposite sign: silence should signal bad news and trade *below* verified low-quality listings.

Figure 1. Non-disclosure price premium around the July 2021 DPE reform

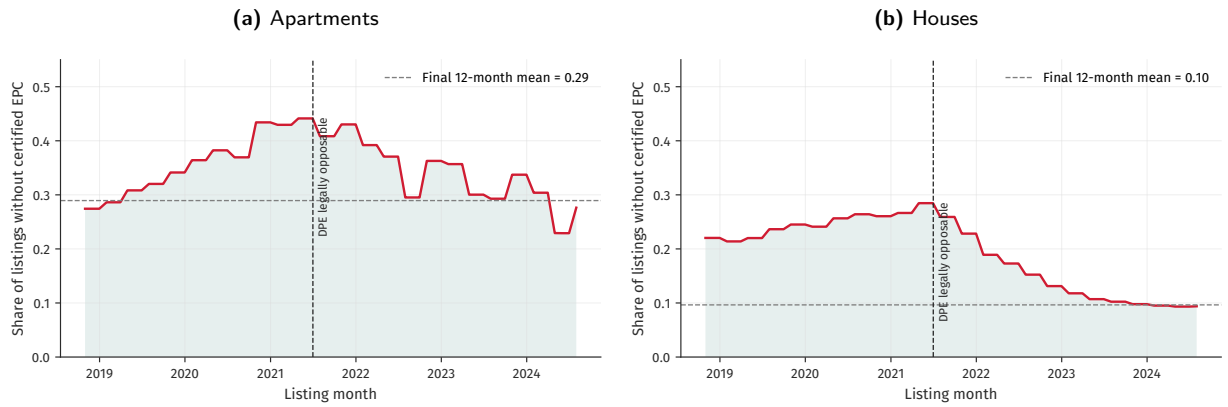


Notes. Monthly volume-weighted log gap between the price index of non-disclosing listings and certified energy-intensive (E/F/G) listings, January 2019 to June 2024. The vertical dashed line marks the legal opposability of the DPE (July 2021). The thick line is a centered three-month moving average. Source: SeLoger residential listings matched against the ADEME registry of certified energy audits; authors' calculations.

2.2 Fact 2: Non-Disclosure Persists Long After Mandatory Disclosure

Classical unraveling further predicts that, once disclosure is mandatory and verifiable, the share of non-disclosing listings should converge rapidly toward zero. The French data reject this prediction in both segments, though with different magnitudes. Figure 2 plots the monthly share of listings without a certified EPC from October 2018 through July 2024. For apartments (panel (a)), the share averages 35.9% before the reform and 33.8% afterward; over the final twelve months of the sample, it stabilizes at 28.9%. For houses (panel (b)), the corresponding numbers are 24.6% pre-reform, 14.4% post-reform, and a final-twelve-month mean of 9.7%. Houses thus display better baseline audit coverage and faster post-reform convergence, but in both segments the share remains structurally bounded away from zero more than three years after disclosure became legally required. Almost three-tenths of apartment listings and one-tenth of house listings still come to market without a certified audit, indicating that a substantial fraction of sellers cannot disclose, because they do not possess a certified audit at the time of listing.

Figure 2. Persistence of non-disclosure after the DPE reform

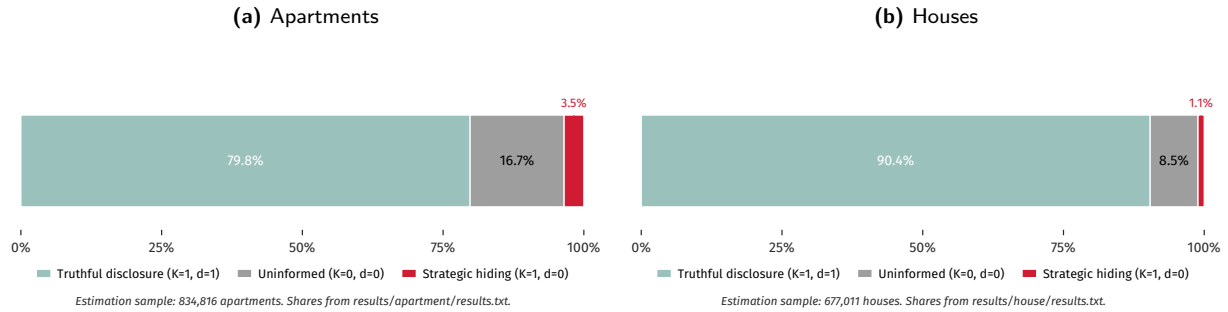


Notes. Monthly share of listings without a certified EPC, October 2018 to July 2024. The horizontal dashed line marks each segment's final-twelve-month mean (0.29 apartments, 0.10 houses). The vertical dashed line marks the July 2021 reform date. Volumes are summed across markets within each segment. Source: SeLoger residential listings matched against the ADEME registry of certified energy audits; authors' calculations.

2.3 Fact 3: Direct Decomposition of the Listing Pool by Information Status

A central methodological contribution of this paper is that, by matching every listing against the ADEME registry, we can directly observe whether a seller is informed ($K = 1$, audit exists) or uninformed ($K = 0$, no audit on record). Crossing this latent information status with the seller's disclosure choice $d \in \{0, 1\}$ yields a four-cell classification of which only three cells are economically distinct: truthful disclosers ($K = 1, d = 1$), strategically hiding sellers ($K = 1, d = 0$), and genuinely uninformed sellers ($K = 0, d = 0$).[1] Figure 3 reports the resulting decomposition for both segments. In apartments (panel (a)), 79.8% of listings are truthful disclosers, 16.7% are genuinely uninformed, and 3.5% are strategically hiding. In houses (panel (b)), the corresponding shares are 90.4%, 8.5%, and 1.1%. Houses systematically display higher baseline disclosure and a smaller strategic-hiding share, but the qualitative structure is identical: a non-trivial mass of strategic hiders coexists with a much larger mass of genuinely uninformed sellers in both segments.

Figure 3. Three-group decomposition of the listing pool



Notes. Shares from the estimation sample of each property type: 834,816 apartment listings and 677,011 house listings, after registry-match cleaning and market-size filtering. Source: SeLoger residential listings matched against the ADEME registry of certified energy audits; authors' calculations.

Two features of this decomposition are central to the analysis. First, the non-disclosing pool ($d = 0$) is mechanically composite in both segments: roughly five-sixths of apartment non-disclosers and seven-eighths of house non-disclosers are genuinely uninformed, while the remainder are strategically concealing. The vast majority of silent listings are silent by constraint, not by choice. Second, the relative weight of these two subgroups varies

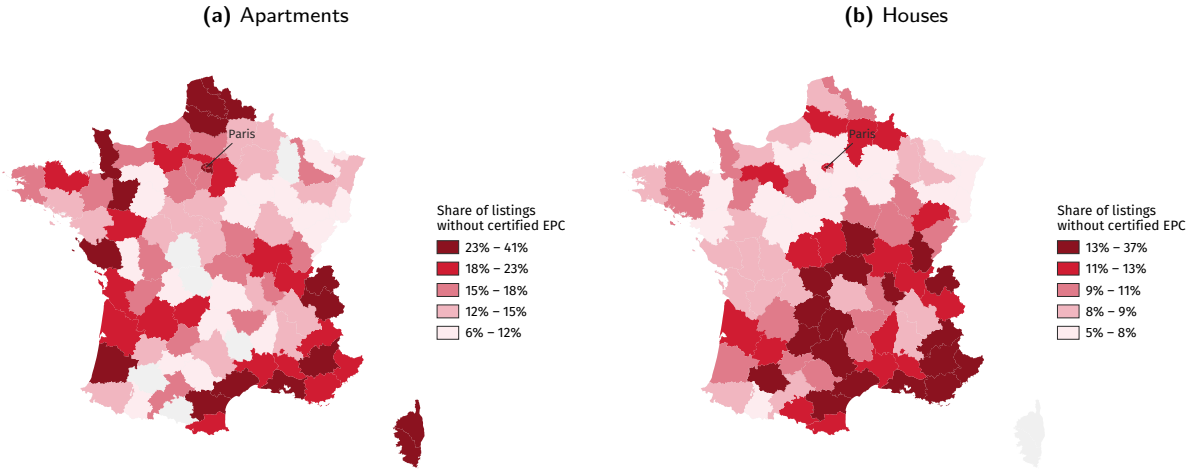
[1] The fourth cell ($K = 0, d = 1$), comprising sellers who display an EPC score without a matched ADEME record, is dominated by registry-match failures rather than genuine misreporting and is excluded from the estimation sample (see Section 5).

substantially across property types and, as we show below, across local markets, with direct implications for the equilibrium informativeness of non-disclosure.

2.4 Fact 4: Substantial Cross-Market Heterogeneity in Non-Disclosure

Fact 4 documents that non-disclosure rates vary sharply across French départements within both property types. Figure 4 maps the apartment (panel (a)) and house (panel (b)) non-disclosure rates at the département level. For apartments, the rate ranges from 6.5% in Aube to 40.9% in Corse-du-Sud, with a cross-département median of 16.5%; high values cluster in southern France, Corsica, and parts of the Parisian region (Seine-Saint-Denis at 32.1%). For houses, the range is 5.4% in Essonne to 37.4% in Paris, with a median of 10.0%; the Paris département is an outlier on the high end, while southern and rural départements lie closer to the metropolitan baseline. The geographic patterns differ across the two segments, reflecting the different local composition of each property type's listing pool, but the underlying message is identical: within-segment cross-département dispersion is large in both cases. This heterogeneity rules out a national-level treatment of the information environment in either segment: any credible structural exercise must allow the depth of audit coverage and the salience of EPC information to vary by local market.

Figure 4. Non-disclosure rate by département, July 2021 to December 2024



Notes. Shares are computed on listings with a valid département code; départements with fewer than 200 listings within each segment are shown in light grey, and overseas départements are omitted. Classification: quantiles ($k = 5$) on metropolitan France, computed separately within each panel so the breakpoints reflect each segment's distribution. Projection: Lambert-93 (EPSG:2154). Source: SeLoger residential listings matched against the ADEME registry of certified energy audits; département boundaries from `france-geojson` (G. David, 2024).

2.5 What the Facts Demand of the Theory

Taken together, the four facts impose tight discipline on any candidate theory of EPC disclosure. Fact 1 (non-disclosure premium) rules out classical full-unraveling models with skeptical buyers, in which silence must trade at or below the worst certified score. Fact 2 (persistence) rules out frictionless implementations in which $\pi = 1$ and disclosure is universal at any positive enforcement cost. Fact 3 (composite non-disclosing pool) requires the theory to recognize that silence aggregates two economically distinct subpopulations whose mass shares (π and the equilibrium hiding rate of informed low-quality sellers) jointly determine the informational content of non-disclosure. Fact 4 (cross-market heterogeneity) rules out a national, scalar treatment of (π, ρ): structural estimation must be conducted market by market. The model developed in the next section is designed to satisfy these four requirements simultaneously.

3 Model

3.1 Motivating Intuition

The two failures interact and reinforce each other. When the informed-seller share and the EPC-sensitive buyer share are both interior, the non-disclosing pool consists of two economically distinct groups: informed

sellers who choose to conceal unfavorable information, and uninformed sellers who cannot disclose because no certified audit exists. Buyers observe only whether a score is present; they cannot distinguish between these groups. A higher uninformed-seller share dilutes the adverse inference buyers draw from non-disclosure, raising the expected quality of the non-disclosing pool and making concealment more attractive for informed low-quality sellers. The result is not merely that non-disclosure persists: silence commands a *premium*. Uninformed sellers do not intend to provide cover; they have no choice. But their mandatory presence in the pool is precisely what makes silence ambiguous and concealment privately rational. Critically, the two parameters are independently movable by policy: the informed-seller share can be raised through audit infrastructure investment, the EPC-sensitive buyer share through energy price reforms or buyer education, and each instrument has distinct, characterizable effects on equilibrium outcomes.

	Fully informed sellers ($\pi = 1$) GM dep. 2	Partially informed sellers ($\pi < 1$)
Buyers do not care ($\rho = 0$)	Disclosure Irrelevant No demand response to disclosure Premium: $\Delta = 0$	Disclosure Irrelevant No demand response to disclosure Premium: $\Delta = 0$
Buyers care ($\rho = 1$) GM dep. 2	Full Unraveling Grossman–Milgrom Silence is fully penalised	Sustained Premium Both channels active ($0 < \rho \leq 1$) $\Delta > 0$

Figure 5. Equilibrium outcomes under seller information and buyer sensitivity heterogeneity

France operates in the bottom-right cell: incomplete audit coverage combined with positive buyer EPC-sensitivity. Matching residential listings to the ADEME registry reveals that 48.8% of apartment listings had no certified energy audit on record at the time of the reform, placing the informed-seller share well below one. On the demand side, the July 2021 reform rendered EPC scores legally opposable, shifting the effective EPC-sensitive buyer share from near zero (when the DPE was merely informational) to a strictly positive level that varies across markets and climates. Even when the EPC-sensitive buyer share is close to one, as estimated for most apartment markets post-reform, the partial-information condition of incomplete audit coverage is sufficient to sustain the premium: as long as a non-trivial share of non-disclosers are genuinely uninformed, buyers cannot fully penalize silence, and the non-disclosure premium persists in equilibrium. Full unraveling requires both conditions to hold simultaneously; France satisfies only one.

3.2 Environment and Timing

A seller owns a property with observable attributes X and an unobservable energy performance level $\theta \in \{0, 1\}$, where $\theta = 1$ denotes high efficiency. Whether the seller knows θ depends on whether she has obtained a certified Diagnostic de Performance Énergétique (DPE) audit prior to listing. With probability $\pi \in (0, 1)$, the seller is *informed* ($K = 1$) and knows θ; with probability $1 - \pi$, she is *uninformed* ($K = 0$) and does not. The parameter π is not a choice variable: it reflects structural features of local audit markets, including the cost and availability of accredited assessors, the time pressure sellers face (relocation, financial distress, inheritance), and the regulatory requirements in force. It is measured directly from the ADEME registry match, as described in Section 5.

The seller simultaneously chooses a disclosure decision $d \in \{0, 1\}$ and a listing price p^{list}. French property law requires that all mandatory diagnostic documents, including the DPE, be appended to the sales agreement at signing (loi ALUR, Article L271-4 CCH), so the disclosure decision and the listing price are set jointly at the time the property is placed on the market. This institutional timing motivates the simultaneous-choice formulation: the seller does not first decide whether to disclose and then price, but sets a complete strategy that internalizes

how the disclosure decision shifts buyer beliefs and thereby affects both the probability of sale and the expected transaction price.

Assumption (No Falsification). *Disclosed EPC scores are verifiable against the ADEME registry. An informed seller with $K = 1$ may choose $d \in \{0, 1\}$, while an uninformed seller with $K = 0$ is constrained to $d = 0$: she cannot disclose what she does not possess. This rules out costless misreporting.*

The key friction is that buyers observe (d, p^{list}, X) but not the seller's information status K: they cannot distinguish a seller who chose not to disclose from one who could not. A buyer then arrives and decides whether to purchase at p^{list}. With probability $\rho \in (0, 1)$, the buyer is *EPC-sensitive* and incorporates energy performance into her valuation; with probability $1-\rho$, the buyer is *EPC-insensitive* and conditions her bid solely on observable attributes. Sellers cannot observe buyer type ex ante and optimize against expected demand. The parameter ρ is market-specific, reflecting the local composition of the buyer pool, and is identified from the differential price response to the July 2021 reform across markets, as detailed in Section 6.

We provide a visual representation of the model dynamics and timing in Figure 6; a fuller schematic overview appears later in Figure 12.

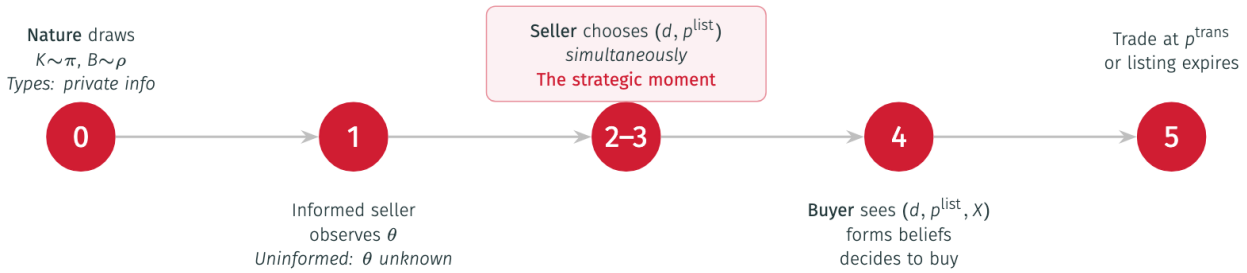


Figure 6. Model timing

3.3 Payoffs and Hiding Costs

In French residential markets, listing prices serve as anchors for negotiation rather than posted take-it-or-leave-it offers: the transaction price conditional on a sale typically falls below the listing price by a margin that depends on local market conditions and property characteristics. Buyer arrival and match quality are stochastic, so the seller faces uncertainty over both whether a sale occurs and, conditional on a sale, at what price. The seller's problem therefore separates into a transaction probability Γ and an expected transaction price ϕ, each depending on the listing price and disclosure decision:

$$\max_{\{d,\, p^{\text{list}}\}} \Pi = \underbrace{\mathbb{E}\left[p^{\text{trans}} \mid d, p^{\text{list}}, X\right]}_{\phi(p^{\text{list}})} \cdot \underbrace{\Pr(T{=}1 \mid d, p^{\text{list}}, X)}_{\Gamma(p^{\text{list}})} - C(\theta, d) \qquad \text{(Eq. 1)}$$

where $\phi(p^{\text{list}}) = \gamma_0 + \gamma_1 \cdot p^{\text{list}} + \sum_n \gamma_n X_n$ is the expected transaction price conditional on a sale occurring, and $C(\theta, d) = \mathbf{1}[d = 0] \cdot (c_0 + \theta c_1)$ is the hiding cost, paid only when the seller withholds information.

The hiding cost has two components. The term $c_0 \geq 0$ applies to all non-disclosers, regardless of quality. In the French institutional context, c_0 captures the regulatory burden imposed by loi ALUR (legal exposure if a buyer subsequently discovers a concealed audit via the public ADEME registry, notaire liability checks, and the informational ambiguity that prolongs negotiation and increases sale failure risk). The term $c_1 \geq 0$ is an additional opportunity cost borne exclusively by high-quality sellers ($\theta = 1$) who choose to hide: by not disclosing, they forgo the revenue premium that verified high quality would command from EPC-sensitive buyers. This asymmetry, $c_1 > 0$, is economically decisive: it means that hiding is strictly cheaper for low-quality sellers than for high-quality ones, so the net gain from concealment is larger for $\theta = 0$ sellers at any given pool belief. This differential is the sorting pressure that drives full separation of high-quality sellers in equilibrium (Corollary 4.1). Hiding costs are paid unconditionally on sale: they represent compliance burdens and reputational exposure that arise at listing time, not at closing.

3.4 Buyer Valuations and Demand

The partition of buyers into EPC-sensitive and EPC-insensitive types reflects a well-documented empirical regularity: willingness to pay for energy efficiency varies sharply with how much the buyer will bear the ongoing energy costs. Primary-residence buyers in northern France with long heating seasons, and investor-buyers who face the impending F/G rental ban under the *loi Climat et Résilience*, have strong incentives to incorporate EPC scores into their bids. Vacation-home buyers who occupy the property a few weeks per year, and buyers in mild climates, may rationally discount energy performance. This heterogeneity determines ρ, the share of the local buyer pool that is EPC-sensitive.

EPC-sensitive buyers ($B = 1$) condition their valuation on energy performance:

$$v_1(\theta, d) = \begin{cases} \beta_0 + \beta_1 \cdot \theta + \sum_n \beta_n X_n & \text{if } d = 1 \text{ (verified quality)}, \\ \beta_0 + \beta_1 \cdot \mathbb{E}[\theta \mid d{=}0] + \sum_n \beta_n \cdot X_n & \text{if } d = 0 \text{ (inferred quality)}. \end{cases} \tag{Eq. 2}$$

We adopt the benchmark $\alpha_k = \beta_k$: buyers value quality equally whether it is certified or inferred. This is a conservative assumption in a precise sense: under the alternative $\alpha_k < \beta_k$ (a distrust discount on unverified quality), the non-disclosure premium would be smaller for any given pool composition, so our identified ρ is a lower bound on the true EPC-sensitive buyer share. The non-disclosure premium we document therefore understates what a fully skeptical buyer would generate. EPC-insensitive buyers ($B = 0$) value the property solely on observables: $v_0(X) = \delta_0 + \sum_n \delta_n X_n$. The aggregate transaction probability mixes the two buyer types:

$$\Gamma(p^{\text{list}}) = \rho \cdot G\big(v_1(d) - p^{\text{list}}\big) + (1 - \rho) \cdot G\big(v_0 - p^{\text{list}}\big) \tag{Eq. 3}$$

where $G(\cdot)$ is the logistic CDF governing idiosyncratic taste shocks. The parameter ρ enters the demand function as a weight on the EPC-sensitive buyer valuation: a higher ρ amplifies the revenue difference between a disclosed high-quality listing and a non-disclosing listing, raising the stakes of the disclosure decision for the seller. It is this amplification channel through which ρ governs disclosure incentives. Empirically, ρ is a market-level structural parameter identified from the cross-market variation in the price response to the July 2021 reform, as developed in Section 6.

3.5 Buyer Beliefs and the Non-Disclosing Pool

When observing a non-disclosing listing, buyers cannot determine whether the seller is strategically concealing an unfavorable audit or simply lacks a certified EPC. They form a posterior belief about expected quality using Bayes' rule. The structure of this belief depends on the composition of the non-disclosing pool, which is itself an equilibrium object.

3.5.1 Who Hides?

Each seller is characterized by two attributes: information status $K \in \{0, 1\}$ and quality $\theta \in \{0, 1\}$. We maintain the following assumption throughout.

Assumption 3.1 (Independence) *$K \perp \theta$, with $\Pr(K = 1) = \pi$ and $\Pr(\theta = 1) = \bar{\theta}$.*

Independence is most plausible for transactions driven by exogenous life events (succession following a death, court-ordered liquidation, or divorce settlement), in which legal proceedings impose binding timing constraints that preclude any quality-contingent decision to obtain a certified EPC. In such cases, the seller's information status K is determined by administrative capacity and procedural deadlines rather than the energy performance of the dwelling; a seller under a judicial mandate to liquidate an estate has neither the incentive nor the discretion to condition audit acquisition on θ. In France, forced transactions of this type constitute a substantial share of the resale market, providing a natural mechanism through which $K \perp \theta$ obtains in the data.

Table 1 summarizes disclosure behavior by seller type. Uninformed sellers have no choice: they cannot disclose information they do not possess (No Falsification Assumption). Informed sellers choose strategically, weighing the revenue gain from disclosure against the option of pooling with the non-disclosing mass. We denote by $\sigma_\theta \equiv \Pr(d = 0 \mid K = 1, \theta)$ the equilibrium hiding probability for an informed seller with quality θ.

Table 1. Disclosure Behavior by Seller Type

Type	Mass	Can Disclose?	Hides?	Hiding Mass
Informed, $\theta = 1$	$\pi\bar{\theta}$	Yes	Prob. σ_1	$\pi\bar{\theta}\sigma_1$
Informed, $\theta = 0$	$\pi(1-\bar{\theta})$	Yes	Prob. σ_0	$\pi(1-\bar{\theta})\sigma_0$
Uninformed, $\theta = 1$	$(1-\pi)\bar{\theta}$	No	Always	$(1-\pi)\bar{\theta}$
Uninformed, $\theta = 0$	$(1-\pi)(1-\bar{\theta})$	No	Always	$(1-\pi)(1-\bar{\theta})$

Notes: $\sigma_\theta \equiv \Pr(d = 0 \mid K = 1, \theta)$ denotes the hiding probability for an informed seller with quality θ. Uninformed sellers cannot disclose and hide with probability one.

3.5.2 The Belief Function

Applying Bayes' rule to the pool composition in Table 1, the expected quality among non-disclosers is:

$$e(\sigma_0, \sigma_1) \equiv \Pr(\theta = 1 \mid d = 0) = \frac{\pi\bar{\theta}\sigma_1 + (1-\pi)\bar{\theta}}{\pi\bar{\theta}\sigma_1 + \pi(1-\bar{\theta})\sigma_0 + (1-\pi)} \qquad \text{(Eq. 4)}$$

The numerator counts the high-quality mass in the pool: informed high-quality sellers who choose to hide ($\pi\bar{\theta}\sigma_1$) together with uninformed sellers who happen to have high quality ($(1-\pi)\bar{\theta}$). The denominator is the total non-disclosing mass. The belief e is therefore a weighted average across two economically distinct populations. When uninformed sellers dominate the pool (π small or σ_0 small), non-disclosure looks like a constraint rather than a choice and beliefs remain favorable to the seller. When strategic low-quality hiders dominate (π large and σ_0 large), the pool is contaminated and beliefs deteriorate. This is why unraveling is incomplete: non-disclosure is inherently ambiguous when $\pi < 1$, whereas in classical disclosure theory all sellers are informed and silence is unambiguous bad news.

Lemma 3.1 (Belief Monotonicity) *$e(\sigma_0, \sigma_1)$ is strictly decreasing in σ_0 and strictly increasing in σ_1.*

Proof 3.1 (Proof sketch) *Increasing σ_0 raises the denominator (more low-quality hiders enter the pool) while leaving the numerator unchanged, so e falls. Increasing σ_1 raises the numerator and the denominator by the same term $\pi\bar{\theta}$, and since $e < 1$ the numerator effect dominates, so e rises. The strict monotonicity in σ_0 is the self-correcting force that ensures equilibrium uniqueness in Section 4: as more informed low-quality sellers choose to hide, buyer beliefs worsen, attenuating the benefit of hiding and restoring incentive compatibility.*

4 Equilibrium Analysis

4.1 Disclosure Incentives

The seller's disclosure decision turns on a single comparison: whether the price attainable by withholding the audit result, $p^*_{d=0}$, priced off the pool belief e, exceeds the price attainable by disclosing true quality θ. A low-quality seller ($\theta = 0$) gains from non-disclosure by pooling with uninformed sellers and extracting the associated quality premium; a high-quality seller ($\theta = 1$) loses from non-disclosure because pooling with low-quality hiders depresses the belief e below 1. This asymmetry drives the central result.

Proposition 4.1 (Disclosure Incentives and Seller Sorting) *Under the benchmark $\alpha_k = \beta_k$, a seller of type (K, θ) strictly prefers disclosure if and only if $\theta > e(\sigma^*_0)$. In particular:*

(i) An informed seller with $\theta = 1$ strictly prefers disclosure for any $e(\sigma^*_0) < 1$.

(ii) An informed seller with $\theta = 0$ strictly prefers non-disclosure whenever $e(\sigma^*_0) > 0$.

Proof. See Appendix C. □

Proposition 4.1 reflects a standard adverse-selection asymmetry. For a high-quality seller, disclosure reveals quality exactly, whereas non-disclosure pools her with low-quality hiders; the condition $e(\sigma^*_0) < 1$ is satisfied for any non-degenerate pool and so is always binding. For a low-quality seller, concealment allows her to price off the pool belief $e(\sigma^*_0) > 0$, which is strictly greater than her true quality $\theta = 0$; the gain from pooling is therefore strictly positive

whenever uninformed sellers are present. The additional hiding cost $c_1 > 0$ borne by high-quality concealers reinforces but does not drive this separation under the benchmark $\alpha_k = \beta_k$: the pool-inference argument alone is sufficient to induce full separation of high-quality sellers, with c_1 becoming binding only when $\alpha_k < \beta_k$ (the distrust discount case).

Corollary 4.1 (High-Quality Sellers Always Disclose) *In any equilibrium,* $\sigma_1^* = 0$.

Proof. Follows directly from Proposition 4.1(i). □

Corollary 4.1 is a partial unraveling result: the top of the quality distribution fully separates. An informed high-quality seller gains nothing from ambiguity and is strictly harmed by being pooled with low-quality hiders; she therefore always discloses. The non-disclosing pool is consequently free of high-quality informed sellers in any equilibrium, and the strategic question reduces entirely to the behavior of informed low-quality sellers.

Since $\sigma_1^* = 0$, the equilibrium belief simplifies. Defining $\sigma_0 \equiv \Pr(d = 0 \mid K = 1, \theta = 0)$, the non-disclosing pool consists exclusively of uninformed sellers of both quality types and informed low-quality hiders:

$$e(\sigma_0) = \frac{(1-\pi)\bar{\theta}}{\pi(1-\bar{\theta})\sigma_0 + (1-\pi)} \tag{Eq. 5}$$

The belief $e(\sigma_0)$ is strictly decreasing in σ_0: each additional low-quality hider enters the denominator with weight $\pi(1-\bar{\theta})$ while leaving the numerator unchanged, diluting pool quality and depressing buyer beliefs.

4.2 Equilibrium Characterization

Let $\Delta(e) \equiv \rho \cdot \beta_1 \cdot e$ denote the revenue gain from non-disclosure for a low-quality seller facing pool belief e. Define the net hiding gain:

$$\Phi(\sigma_0) \;\equiv\; \Delta(e(\sigma_0)) - c_0 \;=\; \frac{\rho\,\beta_1\,(1-\pi)\bar{\theta}}{\pi(1-\bar{\theta})\sigma_0 + (1-\pi)} - c_0. \tag{Eq. 6}$$

The function Φ is strictly decreasing in σ_0: as more informed low-quality sellers conceal, pool quality e deteriorates, the gain from hiding falls, and the incentive to conceal dissipates. This monotone feedback delivers a unique equilibrium without reliance on equilibrium refinements.

The three equilibrium regimes follow directly from comparing Φ to zero at the two boundary values: $\Phi(0) = \Delta(\bar{\theta}) - c_0$ measures the net gain when the pool is at its cleanest, and $\Phi(1) = \Delta(e(1)) - c_0$ when it is maximally contaminated.

Proposition 4.2 (Equilibrium Existence and Uniqueness) *There exists a unique equilibrium, characterized as follows:*

(i) If $c_0 > \Delta(\bar{\theta})$*, the unique equilibrium is* Cost-Deterred Hiding*:* $\sigma_0^* = 0$ *and* $e^* = \bar{\theta}$.

(ii) If $c_0 < \Delta(e(1))$*, the unique equilibrium is* Complete Adverse Selection*:* $\sigma_0^* = 1$ *and* $e^* = (1-\pi)\bar{\theta}/(1-\pi\bar{\theta})$.

(iii) If $\Delta(e(1)) \le c_0 \le \Delta(\bar{\theta})$*, the unique equilibrium is* Partial Pooling*:* $\sigma_0^* \in (0,1)$ *is the unique solution to* $\Delta(e(\sigma_0)) = c_0$.

Proof. See Appendix C. □

The three equilibrium regimes are summarized in Table 2.

Neither corner solution fits the data: universal concealment would require the non-disclosure premium to exceed c_0 even against the most contaminated pool, while universal disclosure would require concealment to be unprofitable even against the cleanest. What we observe instead is an interior equilibrium in which some low-quality sellers disclose and the remainder pool with uninformed non-disclosers. Substituting (Eq. 6) into $\Phi(\sigma_0^*) = 0$ yields the closed-form hiding rate:

$$\sigma_0^* = \frac{1}{\pi(1-\bar{\theta})}\left[\frac{\rho\,\beta_1\,(1-\pi)\bar{\theta}}{c_0} - (1-\pi)\right]. \tag{Eq. 7}$$

This expression makes the model's discipline explicit: the equilibrium hiding rate is determined by four structural parameters, and each comparative static is signed by inspection.

Table 2. Equilibrium Regimes

Equilibrium	**Condition**	**Strategy** σ_0^*	**Belief** e^*	**Premium** $p_0^* - p_1^*$
Cost-Deterred Hiding	$c_0 > \Delta(\bar{\theta})$	0	$\bar{\theta}$	$\rho\beta_1\bar{\theta}$
Complete Adverse Selection	$c_0 < \Delta(e(1))$	1	$\frac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}}$	$\rho\beta_1 e^*$
Partial Pooling	$\Delta(e(1)) \leq c_0 \leq \Delta(\bar{\theta})$	$(0,1)$	Solves $\Phi(\sigma_0^*) = 0$	$\rho\beta_1 e(\sigma_0^*)$

A further consequence of the unique equilibrium is a strict ordering of the non-disclosure premia that, at first glance, inverts the natural intuition:

$$p_0^* - p_1^* = \begin{cases} \rho\beta_1\bar{\theta} & \text{Cost-Deterred Hiding } (\sigma_0^* = 0), \\ c_0 & \text{Partial Pooling } (\sigma_0^* \in (0,1)), \\ \rho\beta_1\dfrac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}} & \text{Complete Adverse Selection } (\sigma_0^* = 1), \end{cases} \tag{Eq. 8}$$

with $\rho\beta_1\frac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}} < c_0 < \rho\beta_1\bar{\theta}$. The premium is highest in the regime with zero strategic hiding. When $\sigma_0^* = 0$ the non-disclosing pool is populated entirely by uninformed sellers, buyers' belief reaches its ceiling $\bar{\theta}$, and sellers extract the full rent $\rho\beta_1\bar{\theta}$. As strategic hiding rises, buyers update down, pool quality deteriorates, and the premium shrinks; in *Complete Adverse Selection* it collapses to its minimum. The *Partial Pooling* value cuts through this ordering in a structurally sharp way: the indifference condition $\Delta(e(\sigma_0^*)) = c_0$ locks the equilibrium premium exactly at c_0, invariant to π and ρ within the regime. The observable price gap between non-disclosing and disclosing low-quality sellers is therefore a direct measurement of the structural hiding cost, not a free function of beliefs.

At the *Cost-Deterred Hiding* corner, the hiding cost is prohibitive against even the most favorable pool: non-disclosure carries no adverse inference, and the premium reflects only the quality of genuinely uninformed sellers. At the opposite corner, *Complete Adverse Selection*, the incentive to conceal survives even in the most contaminated pool; every informed low-quality seller hides, buyers cannot distinguish strategic from uninformed non-disclosure, and the premium collapses to its minimum. Neither corner fits the data, which locates the equilibrium in the interior: in *Partial Pooling* the hiding cost is real but not prohibitive, a fraction of low-quality sellers disclose to escape the contaminated pool while the remainder conceal, and the marginal hider breaks even at the unique interior solution. This is the only regime in which π, ρ, and c_0 are jointly identified.

Milgrom (1981) and Grossman (1981) predict full unraveling when concealment is costless: silence is punished by the worst possible belief, so every informed seller with good news discloses until full revelation obtains. In the present model, full unraveling is precluded by two features working in tandem: the uninformed seller fraction $(1-\pi)$ prevents silence from signaling the lowest quality, and $c_0 > 0$ may price sellers out of concealment entirely. When c_0 is large enough to deter hiding even against the cleanest pool, the *Cost-Deterred Hiding* regime obtains, recovering the spirit of Milgrom-Grossman deterrence through a cost channel rather than a belief penalty. Dye (1985) established the foundational insight that partial pooling survives even without a hiding cost, because the uninformed seller mass dilutes the adverse inference from silence. The present model nests this logic but adds structural content: with $c_0 > 0$, the indifference condition anchors the interior premium at c_0 rather than at a free function of beliefs, so the observable non-disclosure price gap directly identifies the hiding cost, a testable restriction absent from Dye's framework. Buyer heterogeneity, parameterized by ρ, scales the gain from concealment uniformly without altering the regime ordering, explaining why partial pooling persists even as the informed seller share rises: a lower fraction of EPC-sensitive buyers offsets the worsening pool, keeping the interior equilibrium viable.

4.3 Comparative Statics

Proposition 4.3 (Comparative Statics) *The following comparative statics hold:*

(i) In the Complete Adverse Selection *equilibrium:* $\partial(p_0^* - p_1^*)/\partial\pi < 0$ *and* $\partial(p_0^* - p_1^*)/\partial\rho > 0$.

(ii) In the Partial Pooling *equilibrium:* $\partial\sigma_0^*/\partial c_0 < 0$, $\partial\sigma_0^*/\partial\rho > 0$, *and* $\partial\sigma_0^*/\partial\pi < 0$.

Proof. See Appendix C. □

Part (i) reflects two distinct transmission channels. The effect of π operates through the *belief channel*: as the share of informed sellers rises, non-disclosure becomes more informative of low quality, e^* falls, and the gain from concealment dissipates. The effect of ρ operates through the *revenue channel*: a larger share of EPC-sensitive buyers widens the price spread between quality levels, raising the value of concealment.

Part (ii) follows from the implicit function theorem applied to $\Phi(\sigma_0^*) = 0$. Since $\Phi'(\sigma_0^*) < 0$, one obtains $\partial\sigma_0^*/\partial c_0 < 0$ directly. The effect $\partial\sigma_0^*/\partial\rho > 0$ follows because a higher ρ increases Δ at any fixed e, shifting Φ upward and raising the interior equilibrium. The effect of π operates through the belief function: a higher π increases the contamination weight $\pi(1-\bar{\theta})$ in the denominator of $e(\sigma_0)$, reducing $\Delta(e(\sigma_0))$ and thereby shifting Φ downward, so $\partial\sigma_0^*/\partial\pi < 0$. The parameters π and c_0 are therefore *policy substitutes*: each reduces strategic hiding through a distinct channel (π through beliefs, c_0 through costs), and their marginal effects on σ_0^* are both negative and quantifiable from the estimated model.

The premium in (Eq. 8) constitutes a transfer from buyers to concealing sellers, but the welfare cost is distinct from the transfer itself. The allocative distortion arises because EPC-sensitive buyers pay a positive premium for assets whose energy performance falls below what the pooled belief implies; were the concealed information transmitted, these buyers would bid less and the surplus-maximizing allocation would obtain. The hiding cost c_0 is incurred by sellers at listing time and is privately borne but socially wasteful: because c_0 does not enter buyer valuations, it dissipates resources without generating any counterpart in buyer surplus, constituting a pure deadweight loss. A formal efficiency characterization of these two inefficiencies (the allocative distortion and the hiding-cost dissipation) and the optimal subsidy that addresses them are developed in Section 8.

5 Data

5.1 Sources and Sample Construction

The empirical analysis draws on three sources matched at the listing level. **Listing records** are sourced from SeLoger, a major French residential real-estate platform covering sales across metropolitan France. For each listing, we observe the listing price, property characteristics (X_i: area, number of rooms, floor, construction era, location), the disclosure indicator $d_i \in \{0, 1\}$, and, when $d_i = 1$, the displayed EPC letter (A through G). **Transaction records** allow us to connect listings to realized sale prices and time-on-market. **The ADEME registry** is the official national database of certified Diagnostic de Performance Énergétique (DPE) audits, maintained by France's environment and energy management agency and reported by accredited third-party assessors. We match listings to the ADEME registry using address, property type, area, and floor, assigning to each listing an indicator $K = 1$ if a certified audit record exists at the time of listing, and $K = 0$ otherwise. For the structural estimation, listings for which no address is available are dropped, as the ADEME match requires a geocoded address; these account for a small share of the total sample.

The full listing sample covers 2021Q3 to 2024Q4, the post-reform period during which EPC disclosure is legally binding and DPE scores are credible to buyers. The apartment sample comprises **843,621 listings** and the house sample **687,878 listings**. For structural estimation we restrict to 2022 to 2024: the H2 2021 period (the six months immediately following the reform) is excluded because the market had not yet fully repriced the DPE (hedonic estimates of $\hat{\beta}_1$ are near zero or negative in that window), yielding **767,000 apartment listings** and **633,000 house listings** for the structural sample. For 51.2% of apartment listings and 40.3% of house listings, the ADEME match yields $K = 1$; the remainder are classified as genuinely uninformed ($K = 0$, no verified audit on record). For identification of market-level price indices, we use the panel of hedonic price indices constructed for Chapter 1, covering **110 markets** (attraction zones, *zones d'emploi*) with monthly observations from October 2018 through July 2024.

5.2 Four-Group Classification

Combining the listing disclosure decision d_i with the ADEME information status K generates four mutually exclusive groups, summarized in Table 3.

Table 3. Classification of listings and sample composition

Group	Interpretation	d_i	K	Apartments	Houses
Truthful disclosers	Informed seller reveals θ_i	1	1	49.3%	37.8%
Strategic hiders	Informed seller conceals θ_i	0	1	**3.5%**	**1.1%**
Genuinely uninformed	No audit exists; forced non-disclosure	0	0	44.9%	59.7%
Non-certifiable discl.	Displays EPC without verified audit	1	0	2.3%	1.4%
Total				843,621	687,878

Two features stand out. First, **strategic hiding is rare but economically meaningful**: 3.5% of apartment listings and 1.1% of house listings come from informed sellers who chose not to disclose. Among informed sellers ($K = 1$), the hiding rate is 6.9% for apartments and 2.7% for houses. Second, **uninformed sellers are common**: 44.9% of apartment listings and 59.7% of house listings are from sellers with no verified audit: these are the pool of non-disclosers who provide involuntary cover for strategic hiders and whose presence is the key friction in the model.

6 Identification

The two structural objects to identify are ρ_m, the market-specific share of EPC-sensitive buyers, and $c_{0,m}$, the seller hiding cost. The structural equation $\Delta_m = \rho_m \cdot \beta_{1,m} \cdot e^*_m$ relates these to observables: Δ_m and $\beta_{1,m}$ are equilibrium price gaps, and e^*_m is directly observed from the ADEME registry. This might appear to make ρ_m point-identified from a single cross-section via $\rho_m = \Delta_m/(\beta_{1,m} \cdot e^*_m)$. The problem is that cross-sectional price gaps reflect *level* differences across markets, not the demand-induced component we need. Any market characteristic that simultaneously shapes the composition of the undisclosed pool and is correlated with buyer EPC-sensitivity would bias a cross-sectional estimate of ρ_m. Purging this confound requires variation that isolates the *change* in price gaps attributable to buyers becoming EPC-sensitive, not their levels.

The July 2021 DPE reform provides this variation, and the two-step approach uses it as follows. In the first step, a difference-in-differences estimator compares price gaps in each market before and after the reform, stripping out market-specific level differences and isolating the reform-induced shift in buyer demand locally. This yields $\hat{\rho}_m = \hat{\Delta}_m/(\hat{\beta}_{1,m} \cdot \hat{e}^*_m)$, where the hat quantities are DiD estimates (post minus pre period means). In the second step, the hiding cost $c_{0,m}$ is recovered from the equilibrium hiding rate $\hat{\sigma}_{0,m}$ via a closed-form inversion of the seller indifference condition (given the equilibrium characterized in Section 3). No additional external variation is required: $\hat{\sigma}_{0,m}$ is directly observed as the fraction of informed low-quality sellers who withhold their EPC, and the inversion is exact. Together, three observables ($\hat{\beta}_{1,m}$, $\hat{\Delta}_m$, $\hat{\sigma}_{0,m}$) and one registry-based input ($\hat{e}^*_m$) map to two structural parameters ($\hat{\rho}_m$, $\hat{c}_{0,m}$), market by market.

6.1 The July 2021 Reform as a Natural Experiment

Before July 2021, the French DPE was a paper exercise. Scores were methodologically inconsistent across assessors, carried no legal weight, and were routinely ignored in negotiations: a buyer who later discovered a disclosed rating was misleading had no legal recourse. The reform of July 1, 2021 changed the regime fundamentally. It standardized the methodology, imposed stricter accreditation requirements on assessors, and, most importantly, rendered disclosed EPC ratings legally opposable: a buyer can now use a disclosed rating as grounds for price renegotiation or legal action. The DPE became, for the first time, a document with contractual force.

This shift is the source of identification: it created a precise, legally documented change in the value of a disclosed EPC score to buyers, with a known implementation date of July 1, 2021. Whether the supply side of the market responded concurrently is a maintained assumption we cannot directly test with the data at hand; we return to it below.[2]

[2]An alternative strategy based on a Bartik instrument (interacting wholesale gas price variation with the pre-reform share of low-efficiency properties per market) fails at the first stage (F-statistic: 1.0 to 1.6). The French *bouclier tarifaire* (energy price shield, October 2021 to 2023) decoupled household energy costs from the TTF gas price spike driven by the Russian invasion of Ukraine,

The identifying assumption is the standard parallel-trends condition: in the absence of the reform, the cross-group price gaps $\beta_{1,m,t}$ (energy-efficient vs. energy-intensive disclosed properties) and $\Delta_{m,t}$ (undisclosed vs. energy-intensive) would have continued on their pre-reform trajectory. The DiD does not require that buyers were entirely inattentive to EPC scores before July 2021; it requires only that their relative pricing of disclosure groups was not already trending. Any post-reform shift in the gaps, relative to that pre-reform baseline, is then attributable to the change in buyer incentives induced by legal opposability.

The pre-trend evidence supports this assumption directly, and more strongly than a single test would. Both $\beta_{1,m,t}$ and $\Delta_{m,t}$ are statistically flat across all markets in the 33 months before the reform (Figure 11). These two gaps measure distinct economic objects: $\beta_{1,m,t}$ captures buyers' willingness to pay for verified high quality, while $\Delta_{m,t}$ reflects the informational wedge created by strategic non-disclosure. They respond to different primitives of the model. A supply-side confound (a shift in π_m, $\bar{\theta}_m$, or seller composition at the reform date) would generically move these two gaps by different amounts and in potentially different directions, since the mechanisms through which supply composition enters each gap differ. The fact that both gaps are jointly flat before July 2021 and both shift upward simultaneously after the reform, at the same markets and on the same calendar date, is difficult to reconcile with a supply-side story. It is exactly what the demand-side interpretation predicts: legal opposability increases the return to EPC quality for sensitive buyers, raising both the price of verified high quality ($\beta_{1,m,t}$) and the information premium extracted by non-disclosers ($\Delta_{m,t}$) at the same moment.

We acknowledge that supply-side stability (the assumption that π_m, $\bar{\theta}_m$, and $c_{0,m}$ did not shift discontinuously at the reform) is maintained rather than directly tested. Our listing data begin in 2021Q3, so pre-reform ADEME match rates are not observable. The timing of audit decisions provides some reassurance: certified DPE audits are commissioned and completed before listing, with a gap of weeks to months, so the audit stock at any listing date reflects decisions made prior to that date. But this is a maintained assumption, and we treat it as such. The consequence of a supply-side shift at the reform is bounded: any resulting bias in $\hat{\rho}_m$ is upward, reinforcing the interpretation of $\hat{\rho}_m$ as an upper bound rather than a point estimate.[3]

6.2 DiD Identification of ρ_m

The identification uses a panel of monthly hedonic price indices constructed by Chapter 1 from listing-level data covering all three disclosure groups. For each market m, group $g \in \{H, L, \emptyset\}$, and month t, the index $\bar{p}^g_{m,t}$ is the predicted log price per m^2 for a reference property conditional on hedonic controls X_i. The H index covers energy-efficient disclosed properties (EPC grades A to D, $d_i = 1$, $K = 1$); the L index covers energy-intensive disclosed properties (grades E to G, $d_i = 1$, $K = 1$); and the $\emptyset$ index covers all non-disclosing listings ($d_i = 0$), a pooled group that blends strategic hiders ($K = 1$) and genuinely uninformed sellers ($K = 0$). By conditioning on observables, the cross-group gap $\bar{p}^g_{m,t} - \bar{p}^{g'}_{m,t}$ isolates the EPC-attributable price wedge net of compositional differences between groups.

From these indices, two time-varying log price gaps are defined:

$$\beta_{1,m,t} = \bar{p}^H_{m,t} - \bar{p}^L_{m,t} \quad \text{(energy-efficiency premium)} \tag{Eq. 9}$$

$$\Delta_{m,t} = \bar{p}^{\emptyset}_{m,t} - \bar{p}^L_{m,t} \quad \text{(non-disclosure premium)} \tag{Eq. 10}$$

The reform-induced shifts in each gap are estimated by differencing period means:

$$\hat{\beta}_{1,m} = \overline{\beta_{1,m,t}}^{\text{post}} - \overline{\beta_{1,m,t}}^{\text{pre}} \tag{Eq. 11}$$

$$\hat{\Delta}_m = \overline{\Delta_{m,t}}^{\text{post}} - \overline{\Delta_{m,t}}^{\text{pre}} \tag{Eq. 12}$$

where the pre-period spans October 2018 through June 2021 (33 months) and the post-period spans July 2021 through July 2024 (37 months). A volume filter requiring at least 20 average monthly transactions per disclosure group per market retains **40 apartment markets** and **100 house markets**; markets with $\hat{\beta}_{1,m} \leq 0$ are excluded from structural inversion.

removing the demand-side variation the instrument was designed to exploit.

[3] The reform also rendered the DPE legally opposable on the demand side, which could have increased sellers' perceived legal exposure from non-disclosure, shifting $c_{0,m}$ upward. If $c_{0,m}$ rose at the reform, $\hat{\Delta}_m$ reflects both the demand-side response and the cost change, causing $\hat{\rho}_m = \hat{\Delta}_m / (\hat{\beta}_{1,m} \cdot \hat{e}^*_m)$ to overstate true buyer sensitivity. This is the same direction of bias as from supply-side instability: both reinforce the upper-bound interpretation.

To validate the parallel pre-trends assumption and characterise the dynamic profile of the reform, the period-mean estimates are complemented with a month-by-month event study. For each market m and relative month k (where $k = 0$ is July 2021), the normalised gaps

$$\tilde{\beta}_{1,m,k} = \beta_{1,m,t(k)} - \beta_{1,m,t(-1)}, \qquad \tilde{\Delta}_{m,k} = \Delta_{m,t(k)} - \Delta_{m,t(-1)} \tag{Eq. 13}$$

are set to zero at $k = -1$ (the last pre-reform month) by construction. Figure 11 plots the cross-market means with 95% confidence intervals over $k \in [-33, +36]$. The pre-reform coefficients ($k < 0$) are statistically indistinguishable from zero for both gaps, confirming parallel pre-trends. The post-reform coefficients increase gradually from $k = 0$, consistent with progressive capitalisation of the legally opposable DPE. The period-mean estimates in Equations (Eq. 11) and (Eq. 12) average these relative-month profiles over the respective windows.

Given these estimates, the structural equation $\Delta_m = \rho_m \cdot \beta_{1,m} \cdot e^*_m$ (Equation (Eq. 8)) implies a closed-form estimator for buyer sensitivity. Since the reform shifts ρ_m while e^*_m is a supply-side object measured directly from the ADEME registry (Section 6.3 below), the DiD isolates the post-reform demand response:

$$\hat{\rho}_m = \frac{\hat{\Delta}_m}{\hat{\beta}_{1,m} \cdot \hat{e}^*_m} \tag{Eq. 14}$$

The constraint $\hat{\rho}_m \leq 1$ binds in the majority of markets. For apartments, the unconstrained ratio exceeds unity in all 40 markets (cross-market mean of approximately 7.7); for houses, 92 of 107 markets with $\hat{\beta}_{1,m} > 0$ are clipped (cross-market mean ratio of approximately 0.58). Two compounding sources drive the excess: composition bias in $\hat{\Delta}_m$, because the non-disclosure price index pools strategic hiders with genuinely uninformed sellers and inflates the numerator; and thin-cell attenuation of $\hat{\beta}_{1,m}$, because sparse disclosure-group cells compress the denominator. These effects are quantitatively consistent with the magnitudes observed, but they preclude point identification of ρ_m in most markets. In clipped markets, $\rho_m = 1$ is maintained as a working assumption, and $\hat{c}_{0,m}$ is interpreted as an upper bound: since the indifference condition $c_{0,m} = \rho_m \cdot \beta_{1,m} \cdot \hat{e}^*_m(\hat{\sigma}_{0,m})$ is strictly increasing in ρ_m, setting $\rho_m = 1$ overstates $c_{0,m}$ relative to any interior value. Point identification is achieved in 8 house markets where the inversion yields an interior solution; these are characterised separately in the Results.

6.3 Measuring Equilibrium Beliefs Directly

Computing $\hat{\rho}_m$ and $\hat{c}_{0,m}$ requires $\hat{e}^*_m$, the expected quality buyers assign to a non-disclosing listing. In standard signalling models this belief must be jointly estimated with the structural parameters. Here it is **directly observable**: the ADEME registry provides the researcher with the same information structure that e^*_m encodes for buyers.

For each market m, the four-group classification of Section 5 yields three sample frequencies: $\hat{\pi}_m$ (the share of listings with $K = 1$, the ADEME match rate); $\hat{\bar{\theta}}_m$ (the share of informed sellers with high quality, $\theta_i = 1$, EPC $\leq$ D); and $\hat{\sigma}_{0,m}$ (the share of informed low-quality sellers, $K = 1$, $\theta_i = 0$, who hide by setting $d_i = 0$). Substituting into the belief function (Equation (Eq. 5)), at $\sigma^*_1 = 0$ per Corollary 4.1):

$$\hat{e}^*_m = \frac{(1 - \hat{\pi}_m)\,\hat{\bar{\theta}}_m}{\hat{\pi}_m\,(1 - \hat{\bar{\theta}}_m)\,\hat{\sigma}_{0,m} + (1 - \hat{\pi}_m)} \tag{Eq. 15}$$

The numerator is the uninformed high-quality mass in the non-disclosing pool; the denominator is the total non-disclosing mass. No model estimation is involved: $\hat{e}^*_m$ is a function of directly observed sample proportions.

6.4 Closed-Form Identification of $c_{0,m}$

One might expect $\hat{c}_{0,m} = \hat{\Delta}_m$, since the equilibrium indifference condition gives $c_{0,m} = \Delta_m$ for a marginal hider. This logic fails because the "undisclosed" category in the price index pools *two heterogeneous groups*: strategic hiders ($K = 1$, $\theta_i = 0$, $d_i = 0$), who are the economically relevant group, and genuinely uninformed sellers ($K = 0$, $d_i = 0$), who constitute approximately 85% of all non-disclosers. $\hat{\Delta}_m$ is therefore a weighted average of the strategic hider premium and the uninformed-seller premium, an upper bound on $c_{0,m}$, not a point estimate.[4]

[4] A listing-level DiD targeting $K = 1$ non-disclosers directly is infeasible: the listing data begin in 2021, providing no pre-reform baseline at the listing level.

The route to $c_{0,m}$ uses the equilibrium hiding rate $\hat{\sigma}_{0,m}$ among informed low-quality sellers ($K = 1$, $\theta_i = 0$). This group is identified precisely by the ADEME match and is immune to contamination from uninformed sellers. In the Partial Pooling regime, the equilibrium indifference condition

$$\Delta(e^*(\hat{\sigma}_{0,m})\,;\,\hat{\rho}_m) = c_{0,m} \tag{Eq. 16}$$

has a unique solution in $c_{0,m}$ for any observed $\hat{\sigma}_{0,m} \in (0, 1)$. Given the first-stage estimate $\hat{\rho}_m$, this yields the closed-form estimator:

$$\hat{c}_{0,m} = \hat{\rho}_m \cdot \hat{\beta}_{1,m} \cdot \hat{e}^*_m(\hat{\sigma}_{0,m}) \tag{Eq. 17}$$

where $\hat{e}^*_m(\hat{\sigma}_{0,m})$ is the belief evaluated at the observed mixing rate. The estimator is exact: one free parameter ($c_{0,m}$) is matched to one moment ($\hat{\sigma}_{0,m}$) via the equilibrium condition. No numerical optimisation is required.

For house markets, the data provide two distinct upper bounds on $c_{0,m}$. The first is the equilibrium inversion under the maintained $\hat{\rho}_m = 1$ assumption:

$$\hat{c}^{\text{inv}}_{0,m} = \hat{\beta}_{1,m} \cdot \hat{e}^*_m(\hat{\sigma}_{0,m}) \tag{Eq. 18}$$

The second is $\hat{\Delta}_m$, the DiD non-disclosure premium, which is an upper bound on $c_{0,m}$ because uninformed sellers (approximately 75% of non-disclosers for houses) inflate the undisclosed pool's average price above what strategic hiders alone would generate. The cross-market means are $\hat{c}^{\text{inv}}_{0,m} = 0.018$ log points and $\hat{\Delta}_m = 0.034$ log points. Internal consistency requires $\hat{c}^{\text{inv}}_{0,m} \leq \hat{\Delta}_m$, which holds in all but a handful of markets. The tighter bound $\hat{c}^{\text{inv}}_{0,m} = 0.018$ is the preferred estimate and is used in the policy counterfactuals of Section 8.

The structural estimates use listing-level data from 2022 to 2024.[5] The estimation sample comprises **40 apartment markets** and **100 house markets**.

Bootstrap standard errors are computed with $B = 500$ within-market resamples. In each draw, listings in market m are resampled with replacement and all market-level moments $(\hat{\pi}_m, \hat{\bar{\theta}}_m, \hat{\sigma}_{0,m})$ are recomputed. The first-stage estimate $\hat{\rho}_m$ is held fixed at its DiD value across bootstrap iterations; only $\hat{\sigma}_{0,m}$ is resampled to propagate sampling uncertainty into $\hat{c}_{0,m}$ via Equation (Eq. 17). The 500 replications yield percentile-based 95% confidence intervals.

7 Results

7.1 Main Estimates

Table 4 reports market-level estimates for both property types. Panel A covers the DiD estimates from the price index panel; Panel B covers the two-step structural estimates; Panel C covers supply-side observables from the ADEME match.

[5] The year 2021 is excluded from the listing-level analysis because the market had not yet fully repriced the reform: the pooled hedonic estimate of $\hat{\beta}_1$ for apartments in H2 2021 is approximately -0.012, indicating the EPC quality premium had not yet turned positive, which renders the structural equation $\Delta_m = \rho_m \cdot \beta_{1,m} \cdot e^*_m$ unreliable for that period.

Table 4. Market-level estimates by property type

	Apartments			Houses		
Parameter	**Mean**	**Std**	**N**	**Mean**	**Std**	**N**
Panel A: DiD estimates from price index panel						
$\hat{\beta}_{1,m}$ (energy-efficiency premium)	1.7%	4.2	43	5.8%	3.8	107
$\hat{\Delta}_m$ (non-disclosure premium, UB)	6.9%	6.2	43	3.4%	4.8	107
Panel B: Two-step structural estimates, pooled 2022 to 2024						
$\hat{\rho}_m$ (buyer EPC-sensitivity)	95.0%	21.8	40	71.8%	43.8	100
$\hat{c}_{0,m}$ (hiding cost)	1.6%	1.3	40	1.8%	1.9	100
Panel C: Supply-side observables from ADEME match						
$\hat{\pi}_m$ (informed-seller share)	51.8%	9.8	40	41.7%	8.7	100
$\hat{\bar{\theta}}_m$ (mean quality $K{=}1$)	26.5%	7.2	40	28.5%	6.8	100
$\hat{\sigma}_{0,m}$ (informed hiding rate)	7.7%	4.8	40	3.7%	3.1	100

Notes. Panel A: market-level DiD estimates from the price index panel (pre: Oct 2018 to Jun 2021; post: Jul 2021 to Jul 2024), volume-filtered at ≥ 20 transactions/month/EPC category. $N = 43$ apartment markets after excluding 25 with $\hat{\beta}_{1,m} \leq 0$; $N = 107$ house markets after excluding 6. $\hat{\Delta}_m$ is an upper bound for $c_{0,m}$ because the undisclosed category pools strategic hiders and uninformed sellers. Panel B: two-step structural estimates using 2022 to 2024 listing data. $\hat{\rho}_m = \hat{\Delta}_m/(\hat{\beta}_{1,m} \cdot \hat{e}^*_m)$ from closed-form DiD inversion, clipped to $[0, 1]$; $\hat{c}_{0,m}$ is recovered by equilibrium inversion from $\hat{\sigma}_{0,m}$ with $\hat{\rho}_m$ maintained at the clipped value (upper bound when $\hat{\rho}_m = 1$, since $c_{0,m} = \rho_m \cdot \hat{\beta}_{1,m} \cdot \hat{e}^*_m$ is increasing in ρ_m). Panel C: supply-side moments computed as sample frequencies from the ADEME match. Means in Panel A and the $\hat{c}_{0,m}$ row of Panel B are percentage changes (computed as $e^{\delta} - 1$, expressed in percent). Share parameters in Panels B and C ($\hat{\rho}_m$, $\hat{\pi}_m$, $\hat{\bar{\theta}}_m$, $\hat{\sigma}_{0,m}$) are in percentage points. Standard-deviation columns are in the same units as the mean (percentage points).

Table 4 yields a coherent and interpretable picture across all three panels. In Panel A, the energy-efficiency premium $\hat{\beta}_{1,m}$ is approximately 1.7% for apartments and 5.8% for houses: disclosing an energy-efficient EPC score commands that price advantage over disclosing an energy-intensive score, after the July 2021 reform. The non-disclosure premium $\hat{\Delta}_m$ (6.9% for apartments and 3.4% for houses) measures how much more non-disclosers earn over low-quality disclosed properties compared with the pre-reform baseline; because the undisclosed pool commingles strategic hiders with genuinely uninformed sellers, this figure is an upper bound on the true concealment cost rather than a direct estimate of it. Panel B translates these reduced-form inputs into seller-side parameters: $\hat{\rho}_m = 95.0\%$ for apartments and 71.8% for houses indicate that nearly all buyers in the average market price energy performance into their offers, while the inverted hiding cost $\hat{c}_{0,m}$ settles at approximately 1.6–1.8% of property value for both types, a strikingly narrow range given how differently apartments and houses are traded. Panel C illuminates the supply side: just over half of apartment sellers (51.8%) and 42% of house sellers hold a certified energy audit at listing ($\hat{\pi}_m$), and among those informed sellers, roughly 27% own a high-quality property ($\hat{\bar{\theta}}_m \approx 26.5$–28.5%). The most striking contrast across property types lies in the strategic-hiding rate $\hat{\sigma}_{0,m}$: 7.7% of informed apartment sellers who hold a bad EPC choose to withhold it, versus only 3.7% for houses, a difference that mirrors the three-fold gap in $\hat{\beta}_{1,m}$, since a larger forgone quality premium makes concealment a costlier choice for house sellers.

7.2 Structural Estimates and Economic Interpretation

The first-stage estimator $\hat{\rho}_m = \hat{\Delta}_m/(\hat{\beta}_{1,m} \cdot \hat{e}^*_m)$, clipped to $[0, 1]$, yields means of 0.950 for apartments and 0.718 for houses. These averages, however, conceal a pattern of boundary saturation. For apartments, 38 of 40 markets reach the $\hat{\rho}_m = 1$ upper boundary; only the 2 full-unraveling markets pull the mean below unity. For houses, 67 of 75 partial-pooling markets also reach the boundary, with 25 full-unraveling markets contributing $\hat{\rho}_m = 0$ and 8 markets achieving interior identification. The boundary reflects a structural feature of the data: the non-disclosure premium $\hat{\Delta}_m$ typically exceeds $\hat{\beta}_{1,m} \cdot \hat{e}^*_m$, so the data are consistent with full buyer EPC-sensitivity but cannot pin down exactly how close to one ρ_m truly is. We maintain $\hat{\rho}_m = 1$ as the working assumption for these markets

and treat $\hat{c}_{0,m}$ as an upper bound on the true hiding cost accordingly.

Under this convention, hiding costs are small in magnitude: 0.016 log points (approximately 1.6% of property value) for apartments and 0.018 log points (1.8%) for houses. The near-identical level across property types suggests that hiding costs are governed primarily by the legal disclosure regime and audit infrastructure, rather than by property-specific characteristics. For house markets, the data additionally provide a second and independent upper bound from the DiD: $\hat{\Delta}_m = 0.034$ log points. The tighter bound from the equilibrium inversion (0.018) lies strictly below this DiD bound, providing a consistency check: the two estimates are compatible with the same underlying $c_{0,m}$.

The contrast between property types is itself informative. Houses have an energy-efficiency premium three times larger than apartments ($\hat{\beta}_{1,m} \approx 0.058$ vs. 0.017 log points), reflecting their larger surface areas and older building stock, which make heating costs directly material to buyers' running expenses. A higher forgone premium makes concealment more expensive for low-quality sellers, and the data reflect this exactly: the hiding rate $\hat{\sigma}_{0,m}$ is half as large for houses (3.7% vs. 7.7%), and the share of strategic hiders is correspondingly lower (1.1% vs. 3.5% of all listings). The non-disclosure premium $\hat{\Delta}_m$ is also lower for houses (0.034 vs. 0.069), since a thinner pool of strategic hiders means buyers' beliefs about withheld quality are less pessimistic. The entire pattern (higher quality premium, lower hiding rate, lower non-disclosure premium) is the qualitative prediction of the model under equilibrium comparative statics across property types.

One further check concerns the gap between the non-disclosure premium $\hat{\Delta}_m$ and the structural hiding cost $\hat{c}_{0,m}$. The raw DiD estimate exceeds the inverted hiding cost by a factor of roughly 4 for apartments and 2 for houses. This systematic gap is not a contradiction: it is what the model predicts. Uninformed sellers (approximately 83% of non-disclosers for apartments and 89% for houses) earn a non-disclosure premium simply by being in the market, pooling with strategic hiders and inflating the undisclosed category's average price above what strategic concealment alone would generate. $\hat{\Delta}_m$ is the market-level average of this pooled premium; $\hat{c}_{0,m}$ is the cost borne by the marginal informed hider, recovered by targeting only the informed-seller hiding rate $\hat{\sigma}_{0,m}$ in the equilibrium inversion. The two objects measure different things, and the fact that $\hat{\Delta}_m$ greatly exceeds $\hat{c}_{0,m}$ is precisely the compositional mechanism the model is designed to capture.

8 Policy Extension: Renovation Subsidies

8.1 Motivation

The core model treats θ as exogenous. From a policy perspective, the key distortion is that informed low-quality sellers can earn rents by pooling with the uninformed instead of upgrading the underlying asset. A seller with $\theta = 0$ faces three options: hide at cost c_0 and capture a share of the non-disclosure premium, disclose her low quality and accept the discount, or renovate at cost $R(X)$ and disclose verified high quality. The wedge that renovation must overcome is the verified-quality premium $\Pi(d{=}1, \theta{=}1) - \Pi(d{=}1, \theta{=}0)$: the price difference between disclosed high and disclosed low quality. Whenever the engineering cost of renovation exceeds this verified-quality premium, the seller declines to invest, and a renovation subsidy is the natural instrument to close the gap.

A subtler point, developed formally below, is that the renovation threshold's apparent dependence on the hiding cost is illusory in the partial-pooling regime. The renovation comparison pits renovate-and-disclose against hiding: an informed low-quality seller never strictly prefers to disclose her low quality when a positive opacity premium exists, since hiding yields the inflated non-disclosure pool revenue at the cost c_0, and that premium-net-of-cost is weakly positive whenever any seller hides in equilibrium. Strategy 1 (disclose-low) is therefore weakly dominated by Strategy 2 (hide), and the threshold $\bar{R} = \Pi(d{=}1, \theta{=}1) - \Pi(d{=}0; e^*) + c_0$ comes from comparing renovation to hiding alone. The $+c_0$ in this formula is misleading: $\Pi(d{=}0; e^*)$ is itself an equilibrium revenue that adjusts to the hiding cost through the equilibrium hiding rate σ_0^*. In partial pooling, the indifference condition $\Pi(d{=}0; e^*) - c_0 = \Pi(d{=}1, \theta{=}0)$ causes the c_0 term to cancel exactly, and the threshold collapses to the verified-quality premium $\Pi(d{=}1, \theta{=}1) - \Pi(d{=}1, \theta{=}0)$. The regulator nevertheless retains a meaningful disclosure-policy lever: *verification production*, that is, the rate at which sellers obtain EPC audits and enter the informed population π. This margin does not affect the marginal seller's renovation threshold either, but it scales the population of sellers subject to that threshold and, over time, converts the housing stock from uninformed to verified status. The three resulting policy instruments (renovation subsidies, hiding-cost enforcement, and verification enforcement) act on three different margins of the seller's problem and are best understood as complements rather than substitutes.

8.2 Extended Seller Problem

We introduce renovation $r \in \{0,1\}$ such that $\theta' = 1$ if $r = 1$ (for a $\theta = 0$ seller). The informed low-quality seller's extended problem is:

$$\max_{\{r,d,p^{\text{list}}\}} \Pi(r,d,p^{\text{list}} \mid \theta, X) = \phi(p^{\text{list}}) \cdot \Gamma(p^{\text{list}} \mid d, \theta') - C(\theta', d) - (R(X,r) - S) \qquad \text{(Eq. 19)}$$

The subsidy $S \geq 0$ reduces the effective renovation cost. Renovating and then hiding is strictly dominated by renovating and disclosing: a seller who upgrades to $\theta = 1$ has no reason to conceal this.

The three strategies available to an informed low-quality seller are summarized in Table 5. The fourth cell, $(r = 1, d = 0)$, is omitted: renovating and then hiding is strictly dominated by renovating and disclosing, since a verified $\theta = 1$ seller earns strictly more by signalling her quality (Corollary 4.1).

Table 5. Strategies for an Informed Low-Quality Seller ($\theta = 0$)

Strategy	Renovate	Disclose	Buyer observes	Gross revenue	Extra cost
1. Disclose $\theta = 0$	No	Yes	$\theta = 0$ verified	$\Pi(d = 1,\, \theta = 0)$	0
2. Hide	**No**	**No**	$e(\sigma_0^*)$ *(pool quality)*	$\Pi(d = 0)$	c_0
3. Renovate & disclose	**Yes**	**Yes**	$\theta = 1$ verified	$\Pi(d = 1,\, \theta = 1)$	$R(X) - S$

Notes: "Extra cost" is measured relative to Strategy 1 (baseline disclosure at $\theta = 0$). The gross revenue terms $\Pi(d, \theta)$ absorb the transaction cost $C(\theta', d)$ from eq. (Eq. 19) and are distinct objects. $e(\sigma_0^*)$ denotes the equilibrium pool quality belief (Eq. 5). The strategy $(r = 1, d = 0)$ is omitted as strictly dominated.[6]

The relevant non-renovation alternative is Strategy 2 (hide), not Strategy 1 (disclose-low). Whenever the non-disclosure pool commands a positive opacity premium $\Delta(e^*) = \rho\beta_1 e^* > 0$, informed low-quality sellers prefer hiding to disclosing low: the gross revenue gain from being in the non-disclosure pool at least covers the hiding cost. Formally, in any equilibrium with $\sigma_0^* > 0$ (Partial Pooling or Complete Adverse Selection), the equilibrium hiding rate satisfies $\Pi(d = 0; e^*) - c_0 \geq \Pi(d = 1, \theta = 0)$, with equality at the partial-pooling indifference. Strategy 1 is therefore weakly dominated by Strategy 2, and the renovation decision reduces to a binary comparison between Strategy 3 (renovate-and-disclose) and Strategy 2 (hide). Strategy 3 dominates Strategy 2 if and only if

$$\Pi(d = 1, \theta = 1) - \big[R(X) - S\big] \;>\; \Pi(d = 0; e^*) - c_0, \qquad \text{(Eq. 20)}$$

which holds if and only if

$$R(X) - S \;\leq\; \underbrace{\Pi(d = 1, \theta = 1) - \Pi(d = 0; e^*)}_{\text{market premium for verified high quality}} + \underbrace{c_0}_{\text{hiding cost avoided}} \;\equiv\; \bar{R}(\pi, \rho, X).$$

$\bar{R}$ is the total private return the market offers an informed low-quality seller for renovating and disclosing: the revenue gain from upgrading out of the non-disclosure pool to verified high-quality status, plus the hiding cost no longer paid. Absent a subsidy, renovation is privately optimal whenever $R(X) \leq \bar{R}$. The minimum incentive-compatible subsidy is the gap $R(X) - \bar{R}$: the transfer required to restore private incentives when the engineering cost exceeds the market reward.

The dependence of $\bar{R}$ on c_0 is the central object for policy. The original formula (Eq. 20) appears to make $\bar{R}$ increase one-for-one with the hiding cost, since the $+c_0$ term enters linearly. This appearance is misleading: $\Pi(d = 0; e^*)$ is itself an equilibrium revenue evaluated at the equilibrium pool quality $e^* = e(\sigma_0^*)$, and the equilibrium hiding rate σ_0^* adjusts to c_0. Once the equilibrium response is taken into account, $\bar{R}$ in the empirically dominant partial-pooling regime is *invariant* to c_0: the apparent dependence is absorbed by the indifference condition. The following lemma makes this precise.

[6] A seller who renovates to $\theta = 1$ has no incentive to conceal: disclosed high quality commands $\Pi(d = 1, \theta = 1) > \Pi(d = 0)$ for any $e(\sigma_0^*) < 1$, incurring the same renovation cost $R(X) - S$ regardless of disclosure. Strategy 3 also dominates Strategy 1 whenever $R(X) - S < \Pi(d = 1, \theta = 1) - \Pi(d = 1, \theta = 0)$; but the policy analysis focuses on the transition from Strategy 2, since the concealment trap binds only for sellers who are already hiding.

8.3 The Renovation Threshold and the Minimum Incentive-Compatible Subsidy

Definition 8.1 (Renovation Threshold) *The renovation threshold $\bar{R}(\pi, \rho, X)$ is the maximum renovation cost at which an informed low-quality seller prefers to renovate-and-disclose over hiding, absent subsidies:*

$$\bar{R}(\pi, \rho, X) \equiv \underbrace{\Pi(d=1, \theta=1) - \Pi(d=0; e^*)}_{\text{Market premium for verified high quality}} + \underbrace{c_0}_{\text{Hiding cost avoided}}. \tag{Eq. 21}$$

Lemma 8.1 (Equilibrium reduction in partial pooling) *In the partial-pooling equilibrium, the indifference condition $\Pi(d=0; e^*) - c_0 = \Pi(d=1, \theta=0)$ implies that the renovation threshold reduces to the verified-quality premium:*

$$\bar{R}^{PP}(\rho, X) = \Pi(d=1, \theta=1) - \Pi(d=1, \theta=0), \tag{Eq. 22}$$

which depends on ρ and X alone and is invariant to c_0 and π. In Complete Adverse Selection ($\sigma_0^* = 1$), the threshold retains the form (Eq. 21) with $e^* = (1-\pi)\bar{\theta}/(1-\pi\bar{\theta})$ locally constant in c_0, so $\partial \bar{R}^{CAS}/\partial c_0 = 1$.

Proof. See Appendix C. □

The partial-pooling reduction (Eq. 22) is the central technical result of this section. It rests on the equilibrium indifference identity: the marginal informed-low seller is, by construction of the regime, exactly indifferent between hiding and disclosing-low, so the hide payoff $\Pi(d=0; e^*) - c_0$ equals the disclose-low payoff $\Pi(d=1, \theta=0)$. Substituting this identity into (Eq. 21) causes the hiding cost to cancel exactly, leaving the verified-quality premium $\Pi(d=1, \theta=1) - \Pi(d=1, \theta=0)$: the revenue gain a seller obtains by upgrading from a verified-low to a verified-high listing.

The economic content of the reduction is that, in the empirically dominant interior regime, the renovation incentive is governed entirely by the price gap between verified high and verified low quality. The non-disclosure premium $\Delta = \rho\beta_1 e^*$, which in partial pooling equals c_0 by the same indifference, plays no role in the renovation calculus: the seller's gross premium Δ for being in the non-disclosure pool and her cost c_0 for hiding cancel because the regime is constructed to make them cancel. Raising the hiding cost shifts the equilibrium hiding rate σ_0^* along the indifference curve (with $\Pi(d=0; e^*)$ adjusting endogenously to preserve indifference) but does not change the renovation threshold. Policymakers cannot "buy" renovation by tightening hiding-cost enforcement in this regime; renovation requires a direct subsidy that improves the renovate-and-disclose payoff relative to the seller's continuation value at low quality.

Definition 8.2 (Minimum incentive-compatible subsidy) *The minimum incentive-compatible subsidy is the smallest transfer making renovation weakly preferred to hiding, for a seller with renovation cost $\hat{R}(X)$, holding equilibrium beliefs fixed:*

$$S^*(\pi, \rho, X) \equiv \max\Big\{\hat{R}(X) - \bar{R}(\pi, \rho, X),\ 0\Big\} \tag{Eq. 23}$$

where $\hat{R}(X)$ denotes the actual renovation cost from engineering estimates or program data.

Remark (S^* is incentive-compatibility, not welfare optimum). The object S^* is the minimum transfer that flips the marginal informed-low seller's decision rule from concealment to renovation, given everyone else's equilibrium behaviour. It is *not* a Ramsey-optimal subsidy. A full welfare problem would maximise

$$\max_S \int_X \mathbf{1}\{\hat{R}(X) \le \bar{R} + S\} \cdot \big[\Delta W(X) - \lambda S\big]\, dF(X),$$

where $\Delta W(X)$ is the social gain per renovation (private surplus change plus any energy-use externality), $\lambda \ge 1$ is the marginal cost of public funds, and F is the within-market distribution of renovation costs. The corner solution $S^* = \hat{R} - \bar{R}$ emerges only in the limit of degenerate F, $\lambda = 1$, and zero externality. Throughout this section we treat S^* as the operative *positive* object (the smallest subsidy that induces a given seller to renovate) and leave the full Ramsey characterisation to future work.

The renovation threshold $\bar{R}$ is the market's implicit valuation of quality information from the seller's perspective. In the partial-pooling equilibrium it equals the verified-quality premium $\Pi(d=1, \theta=1) - \Pi(d=1, \theta=0)$, the revenue gain from a buyer-verifiable upgrade from low to high quality. In Complete Adverse Selection it

retains the original definition (Eq. 21) with the equilibrium pool quality at the corner. When actual renovation costs fall below the threshold, the market already rewards quality revelation enough to make private renovation worthwhile. When $\hat{R}(X) > \bar{R}$, the market premium for renovation is insufficient and targeted public support is required. The minimum incentive-compatible subsidy S^* is precisely the difference: it fills the gap between what the market offers and what renovation costs, doing no more than necessary to tip the seller's calculus toward quality improvement.

8.4 Comparative Statics: Renovation and Disclosure Operate on Different Margins

Proposition 8.1 (Subsidy comparative statics by regime) *The minimum incentive-compatible subsidy $S^*(\pi, \rho, X)$ satisfies $\partial S^*/\partial \hat{R}(X) = \mathbf{1}\{\hat{R}(X) > \bar{R}\}$ throughout, together with the following regime-conditional comparative statics on the renovation threshold $\bar{R}$:*

(i) **Partial Pooling** *(the empirically relevant interior regime): $\partial \bar{R}/\partial c_0 = 0$, $\partial \bar{R}/\partial \pi = 0$, and $\partial \bar{R}/\partial \rho > 0$. Hence*

$$\frac{\partial S^*}{\partial c_0} = 0, \qquad \frac{\partial S^*}{\partial \pi} = 0, \qquad \frac{\partial S^*}{\partial \rho} \leq 0 \qquad \text{when } \hat{R}(X) > \bar{R}.$$

(ii) **Complete Adverse Selection** *($\sigma_0^* = 1$, corner): $\partial \bar{R}/\partial c_0 = 1$, $\partial \bar{R}/\partial \pi > 0$, $\partial \bar{R}/\partial \rho > 0$. Hence $\partial S^*/\partial c_0 = -1$ and $\partial S^*/\partial \pi \leq 0$ when $\hat{R} > \bar{R}$.*

Proof. See Appendix C. □

The threshold result of Proposition 8.1 should not be read as evidence that disclosure policy is irrelevant for renovation outcomes. It is irrelevant only for one specific decision: the marginal informed low-quality seller's choice between concealment and renovation, holding her informed status fixed. The broader policy problem involves three distinguishable instruments acting on three distinguishable margins.

The first is the intensive margin of renovation propensity. The subsidy S acts directly on the seller's renovation calculus through the inequality $\hat{R}(X) - S \leq \bar{R}$. Conditional on being informed and low quality, the probability of renovation is governed entirely by S and by the cross-property distribution of $\hat{R}(X)$.

The second is the composition of disclosure decisions among the informed. The hiding cost c_0, raised through fines, audits of suspected concealers, or strengthened legal exposure under loi ALUR, acts on the share of informed-low sellers who hide versus disclose-low. By Proposition 4.3(ii), $\partial \sigma_0^*/\partial c_0 < 0$ within partial pooling: tightening hiding penalties shrinks the strategic-hiding share. By Lemma 8.1, however, this leaves the renovation threshold unchanged in partial pooling: the equilibrium pool revenue $\Pi(d = 0; e^*)$ adjusts to keep the hider exactly indifferent with the disclose-low alternative, so the $+c_0$ in (Eq. 21) is exactly offset by the equilibrium movement of $\Pi(d = 0; e^*)$. Only in the Complete Adverse Selection corner, where $\sigma_0^* = 1$ is at the boundary and $\Pi(d = 0; e^*)$ is locally constant in c_0, does c_0 enter the threshold and the textbook substitutability $\partial S^*/\partial c_0 = -1$ obtain.

The third is the extensive margin of verification production. Mandatory EPC enforcement, audit infrastructure investment, and inspection of cadastral records raise π, the share of sellers who are informed about their property's energy performance, which is the principal long-run instrument in France: every transaction since 2007 in principle triggers an EPC, and the ADEME registry grows monotonically over time. The effect on aggregate renovation is direct and first-order:

$$\text{Renovation flow} = \underbrace{\pi}_{\text{informed share}} \cdot \underbrace{(1 - \bar{\theta})}_{\text{low among informed}} \cdot \underbrace{F_R\big(\bar{R}(\rho, X) + S\big)}_{\text{share with } \hat{R} \leq \bar{R} + S}, \qquad \text{(Eq. 24)}$$

where F_R is the within-market distribution of engineering renovation costs. Raising π scales the renovation base linearly even when the per-seller threshold is invariant, because additional sellers become informed and enter the strategic disclosure problem in the first place. Each newly verified low-quality property is a potential subsidy recipient that previously was not.

There is also a stock-evolution effect that operates only along the verification-production margin. As π rises and audits accumulate, the housing stock becomes increasingly characterised by verified energy labels. Buyers can condition prices on observed θ for a growing share of the market; the uninformed pool that sustains the

non-disclosure premium shrinks; and the equilibrium gradually drifts from partial pooling toward cost-deterred hiding as the gain from concealing low quality erodes. Renovations undertaken today, once disclosed at the next sale, become permanent reclassifications from $\theta = 0$ to $\theta = 1$ in the verified stock. The static threshold analysis above takes the share of informed sellers as given; the verification enforcement margin is precisely what makes this share endogenous to policy over the medium run.

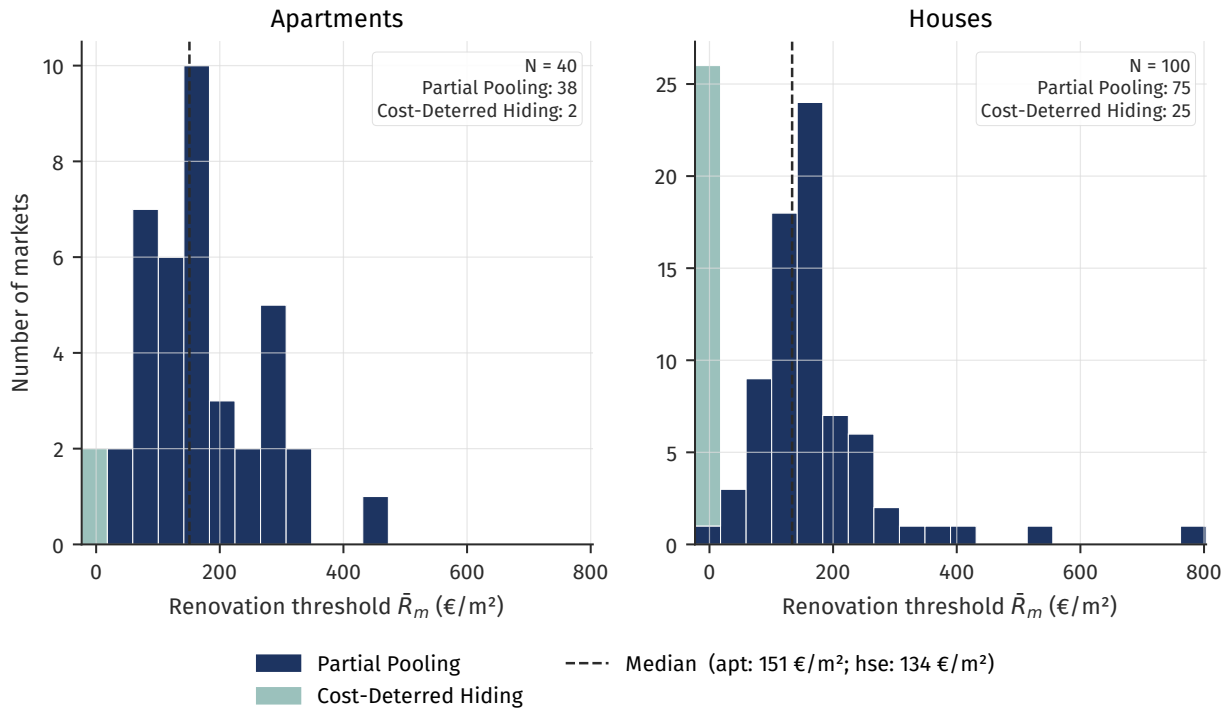
The minimum incentive-compatible subsidy $S^*(\pi, \rho, X)$ measures how much transfer it takes to flip the renovation decision of a single informed low-quality seller, given the equilibrium pool composition, and the intensive margin alone determines that object. Hiding-cost enforcement does not substitute for the subsidy in the empirically dominant partial-pooling regime: at the indifference margin, the apparent $+c_0$ in the threshold is exactly cancelled by the equilibrium movement of $\Pi(d = 0; e^*)$, and the renovation incentive collapses to the verified-quality premium. Verification production is genuinely complementary: π does not change the threshold either, but it directly expands the population subject to it and dynamically converts the housing stock toward verified status. Disclosure enforcement and renovation subsidies are therefore complements that scale each other's reach, not substitutes.

A uniform national subsidy is generically inefficient. The threshold $\bar{R}(\pi, \rho, X)$ varies across markets through ρ (the share of EPC-sensitive buyers) and through property characteristics X, even in partial pooling where π and c_0 drop out of the marginal threshold. A uniform national subsidy therefore necessarily over-subsidises markets with high ρ (where the verified-quality premium is already large) and under-subsidises markets with low ρ (where it is small). The verification-production margin reinforces this: at any given S, the renovation flow (Eq. 24) also scales with the local π, so subsidies and audit-infrastructure investment should be co-targeted on markets where both π is low and ρ delivers a sufficient verified-quality premium to make subsidised renovation cost-effective. Efficient policy requires targeting on observable proxies for market composition: local climate and energy-cost exposure for ρ, audit infrastructure (notaire density, ADEME registry coverage) for π, and property-level engineering estimates for $\hat{R}(X)$.

8.5 Renovation Subsidies in French Housing Markets: Empirical Magnitudes and Spatial Heterogeneity

In the partial-pooling regime, Lemma 8.1 reduces the renovation threshold $\bar{R}^{\text{PP}}$ to the verified-quality premium $\Pi(d{=}1, \theta{=}1) - \Pi(d{=}1, \theta{=}0)$, with c_0 and π cancelling through equilibrium indifference. Because β_1 measures the quality premium in log-point terms against a local price-per-m^2 base, the nominal threshold $\bar{R}_m$ scales with the local price level: high-price urban markets have the largest verified-quality premia in absolute terms, even when the proportional premium β_1 is similar across locations. Across the 40 estimable apartment markets, $\bar{R}_m$ averages €165/m^2 (median €151/m^2); across the 100 house markets, the mean is €128/m^2 and the median €134/m^2 (Figure 7). The two Cost-Deterred Hiding apartment markets and the 25 CDH house markets sit at the upper tail of their respective distributions: in CDH, $\sigma_0^* = 0$ so the equilibrium pool is uncontaminated and pool revenue is at its maximum, yielding a large verified-quality premium. The PP markets cluster more narrowly around their medians, reflecting the equilibrium compression that the indifference condition imposes on pool revenue. Cross-market dispersion is nonetheless substantial: apartment thresholds range from roughly €80 to €350/m^2, so markets at the upper tail of the distribution are intrinsically more amenable to privately funded renovation than those at the lower tail.

Figure 7. Distribution of the renovation threshold $\bar{R}_m$ across markets

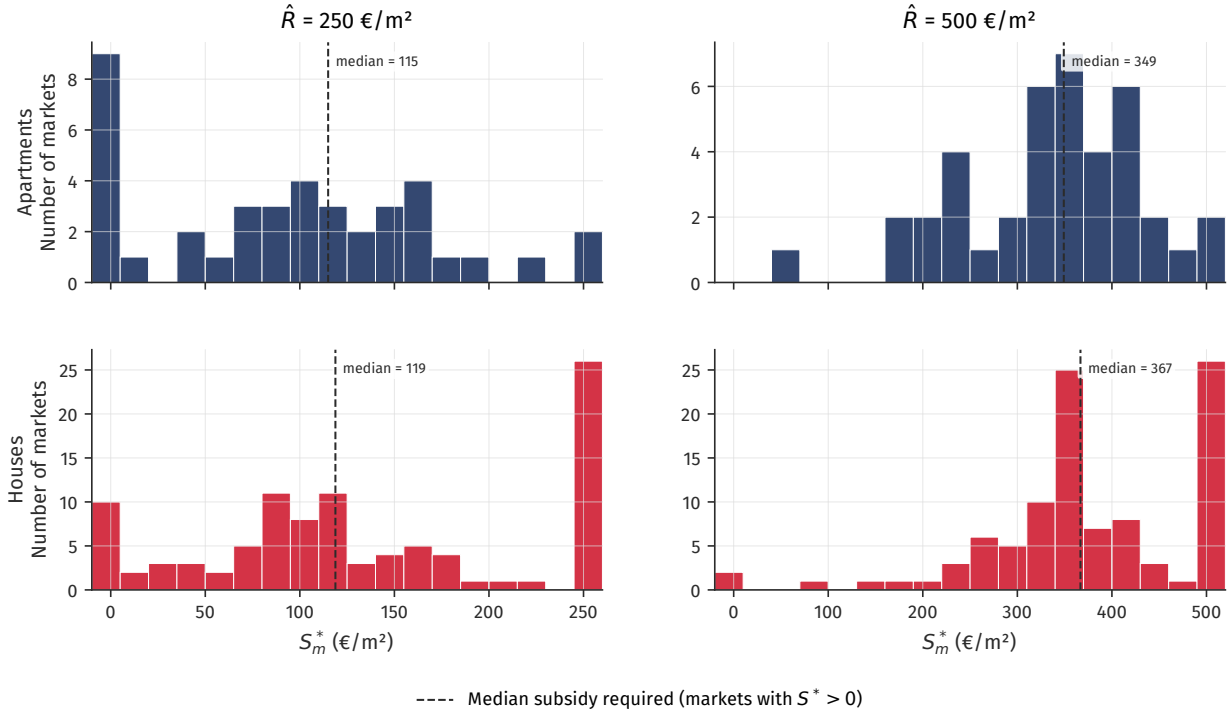


Notes. Each panel shows the cross-market distribution of $\bar{R}_m$ (euros per m^2) for one property type, with stacks coloured by equilibrium regime: Partial Pooling (PP) and Cost-Deterred Hiding (CDH). Dashed vertical lines mark within-panel medians. In PP, the threshold equals the verified-quality premium $\Pi(d{=}1,\theta{=}1) - \Pi(d{=}1,\theta{=}0)$ by Lemma 8.1. In CDH, the full formula (Eq. 21) applies and the pool is uncontaminated ($\sigma_0^* = 0$).

The minimum incentive-compatible subsidy is $S_m^* = \max\{\hat{R} - \bar{R}_m, 0\}$, evaluated here at two renovation-cost benchmarks: €250/m^2 (a light, "geste par geste" intervention such as roof or wall insulation) and €500/m^2 (a medium, two-measure upgrade), drawing on MaPrimeRénov' programme documentation.[7] At $\hat{R} = 250$, 31 of 40 apartment markets (77.5%) require a positive subsidy; the median S_m^* conditional on $S_m^* > 0$ is €99/m^2, or approximately €6,048 per unit at the median apartment area of 63 m^2. For houses, 91 of 100 markets (91%) require support, with a conditional median of €116/m^2 (approximately €13,287 per unit at 113 m^2). By $\hat{R} = 500$, every apartment market requires a subsidy and 98 of 100 house markets do likewise; the conditional medians rise to €349/m^2 (apartments, €22,192 per unit) and €366/m^2 (houses, €42,060 per unit). The cost-sharing ratio $S_m^*/\hat{R}$ is the natural decision variable for the regulator: at the light benchmark it stands at roughly 40% for the median apartment market and 46% for the median house market; at the medium benchmark both rise to approximately 70%. This non-linear escalation in required cost-sharing as intervention depth increases has a direct programme-design implication: a commitment to a fixed cost-sharing rate is more defensible than a commitment to a fixed euro amount, because it adjusts automatically to the depth of the intervention and to local market conditions. The absolute per-unit gap between apartments and houses reflects floor area rather than any structural difference in the per-m^2 economics: the policy-relevant unit is euros per square metre, and a programme denominated in euros per unit mechanically over-targets houses (median 113 m^2) relative to apartments (median 63 m^2) for any given social-welfare weight. Figure 8 presents the full cross-market distribution of S_m^* at both benchmarks; Figure 9 decomposes it by equilibrium regime.

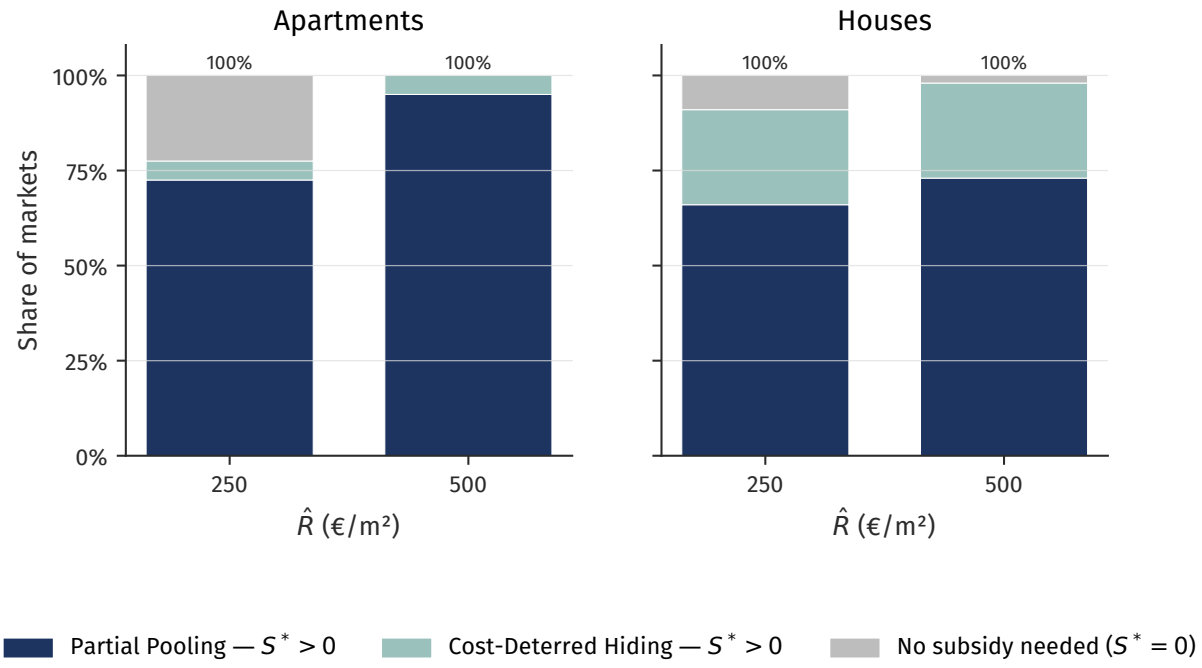
[7] MaPrimeRénov' distinguishes "geste par geste" interventions (insulation of walls or roof, costing roughly €100–400/m^2 of treated area) from comprehensive "rénovation d'ampleur" operations (€700–1,200/m^2 or more); the €500/m^2 benchmark corresponds broadly to a two-measure upgrade.

Figure 8. Distribution of the minimum incentive-compatible subsidy S_m^* across markets



Notes. Rows correspond to property type (apartments, houses); columns to the renovation-cost benchmark $\hat{R} \in \{250, 500\}$ €/m². Each cell shows the cross-market density of S_m^* (euros per m²). The dashed vertical line within each cell marks the median conditional on $S_m^* > 0$. Markets with $S_m^* = 0$, for which renovation is privately optimal without intervention, appear in the left spike. The benchmark $\hat{R} = 250$ €/m² corresponds to a light, "geste par geste" intervention; $\hat{R} = 500$ €/m² to a medium, two-measure upgrade.

Figure 9. Share of markets needing a subsidy, by regime and renovation-cost benchmark



Notes. Each bar sums to 100% of markets in its (property type, $\hat{R}$) cell. The coloured stacks distinguish Partial Pooling (PP) and Cost-Deterred Hiding (CDH) markets that require a positive subsidy; the grey residual labels markets with $S_m^* = 0$, for which the verified-quality premium already exceeds the renovation cost. Benchmarks shown: $\hat{R} \in \{250, 500\}$ €/m². In PP, the threshold $\bar{R}_m$ is invariant to c_0 (Lemma 8.1); in CDH, $\bar{R}_m$ absorbs c_0 and is consequently larger, so CDH markets are over-represented among those needing no subsidy.

The regime decomposition is informative for understanding the structural mechanism, though it is secondary to spatial targeting for programme design. In PP, the threshold is invariant to c_0 by Lemma 8.1, so cross-market dispersion in S_m^* is driven entirely by variation in β_1 and local prices. CDH markets retain the full formula (Eq. 21) and carry larger thresholds; they are consequently over-represented in the "no subsidy needed" category at $\hat{R} = 250$ and require smaller subsidies at $\hat{R} = 500$ than their sample share would suggest. This CDH group is a minority (2 apartment markets, 25 house markets) and is, by construction, the segment where the private renovation incentive is already strongest. The policy challenge lies in the PP majority, where the renovation threshold is pinned at the verified-quality premium and public intervention must bridge the gap between that premium and the renovation cost.

A uniform national subsidy is inefficient because the cost-sharing requirement varies substantially across markets. Figure 10 maps $S_m^*/\hat{R}$, the share of the renovation invoice the regulator must cover, at the two cost benchmarks; bubble size encodes the local verified-quality premium $\bar{R}_m$. The support of the cost-sharing distribution is wide. At $\hat{R} = 250$ €/m², 22.5% of apartment markets and 9% of house markets need no subsidy at all, while markets at the upper end of the distribution require cost-shares approaching 80%. A flat per-m² grant set at the cross-market mean would simultaneously over-pay the lower tail of this distribution, where the private incentive already suffices, and under-pay the upper tail, where renovations remain unfunded. Both forms of misallocation are costly: the former through fiscal waste on infra-marginal markets, the latter through forgone renovations in the markets where the gap is largest. This cross-market dispersion is systematic rather than idiosyncratic, since the cost-sharing ratio is mechanically low where $\bar{R}_m$ is high, and $\bar{R}_m$ is high where local prices per m² are large and buyer EPC-sensitivity ρ_m is strong. These conditions hold in the high-price urban markets of metropolitan France, which carry the largest bubbles on the maps; the cost-sharing ratio is correspondingly higher in lower-price peripheral and rural markets, where small bubbles signal that the verified-quality premium delivers only a modest fraction of the renovation cost. An efficient programme would differentiate the per-m² subsidy along these structural axes rather than apply a uniform amount, channelling more public expenditure to markets where the private renovation

gap is largest. The bubble-size dimension also points to the co-targeting logic of §8: markets with small bubbles are typically markets with low verification penetration π, combining a small per-renovation base with a small eligible population. Effective deployment therefore requires simultaneously raising π through audit-infrastructure investment and concentrating the cost-sharing transfer on the markets where both margins are most constrained.

Figure 10. Spatial distribution of the minimum incentive-compatible subsidy across metropolitan France



Notes. Top row: $\hat{R} = 250$ €/m^2 (light intervention). Bottom row: $\hat{R} = 500$ €/m^2 (medium intervention). Left panels show apartment markets; right panels show house markets. Projection: Lambert-93 (EPSG:2154); the bar in the lower-left of each panel measures 100 km. Bubble colour encodes the cost-sharing ratio $S_m^*/\hat{R}$, the share of the renovation cost the regulator must cover; bins are fixed at {0–20%, 20–40%, 40–60%, 60–80%, 80–100%} so colours are directly comparable across cost benchmarks. Bubble size encodes the renovation threshold $\bar{R}_m$ (verified-quality premium in euros per m^2), pooled across both property types and both benchmarks so sizes are comparable across all four panels. Markets with $S_m^* = 0$ are displayed as small uniform grey dots.

Aggregating S_m^* over all listings yields upper-bound programme costs under the counterfactual that every eligible listing is renovated. For apartments (641,987 listings), the aggregate is €6.4 billion at $\hat{R} = 250$ and €15.7 billion at $\hat{R} = 500$. For houses (508,732 listings), the corresponding figures are €9.1 billion and €23.0 billion. The combined totals are €15.5 billion at the light benchmark and €38.7 billion at the medium benchmark. MaPrimeRénov's current annual budget is approximately €4–5 billion; a medium-intervention national rollout would therefore require a 3–8× scaling of current spending. That multiplier is the headline fiscal fact: it implies that any credible decarbonisation programme targeting the housing stock must either substantially expand the fiscal envelope or rely heavily on the cost-sharing and co-targeting logic developed above to concentrate expenditure where the private-renovation gap is largest. These estimates should be read as order-of-magnitude upper bounds. The per-listing aggregation abstracts from the extensive-margin participation decision of sellers, and the realised fiscal cost under any actual programme would be smaller. More fundamentally, Remark 8.3 establishes that S_m^* is an incentive-compatibility object rather than a Ramsey-optimal transfer: with a shadow cost of public funds $\lambda > 1$, the regulator will set the operative subsidy below S_m^*, trading off the verified-quality-premium gap against the deadweight cost of taxation. The maps and distributions above identify where market failures are largest; they are properly read as identifying the targets of maximum social return per euro of public expenditure, not as direct programme prescriptions.

9 Conclusion

We have developed a theory of disclosure failure grounded in two empirically motivated departures from the classical framework: sellers may be uninformed, and buyers may be heterogeneous in their valuation of quality information. These features jointly sustain non-disclosure premia in equilibrium: silence is ambiguous rather than damning, and low-quality sellers benefit from pooling with genuinely uninformed sellers who provide involuntary cover. The interaction between the two departures is essential: buyer heterogeneity ensures that concealment is profitable even when beliefs are pessimistic; seller information heterogeneity ensures that non-disclosure is insufficiently revealing to discipline strategic hiding.

The key parameters governing market outcomes are π (the share of informed sellers) and ρ (the share of quality-sensitive buyers). Full unraveling (the classical benchmark) emerges only when both are high. For intermediate values, partial separation with persistent non-disclosure premia is the generic outcome, and the premium is structurally determined by the uninformed-seller share: it measures, in equilibrium prices, how much ambiguity is left in the market's information environment.

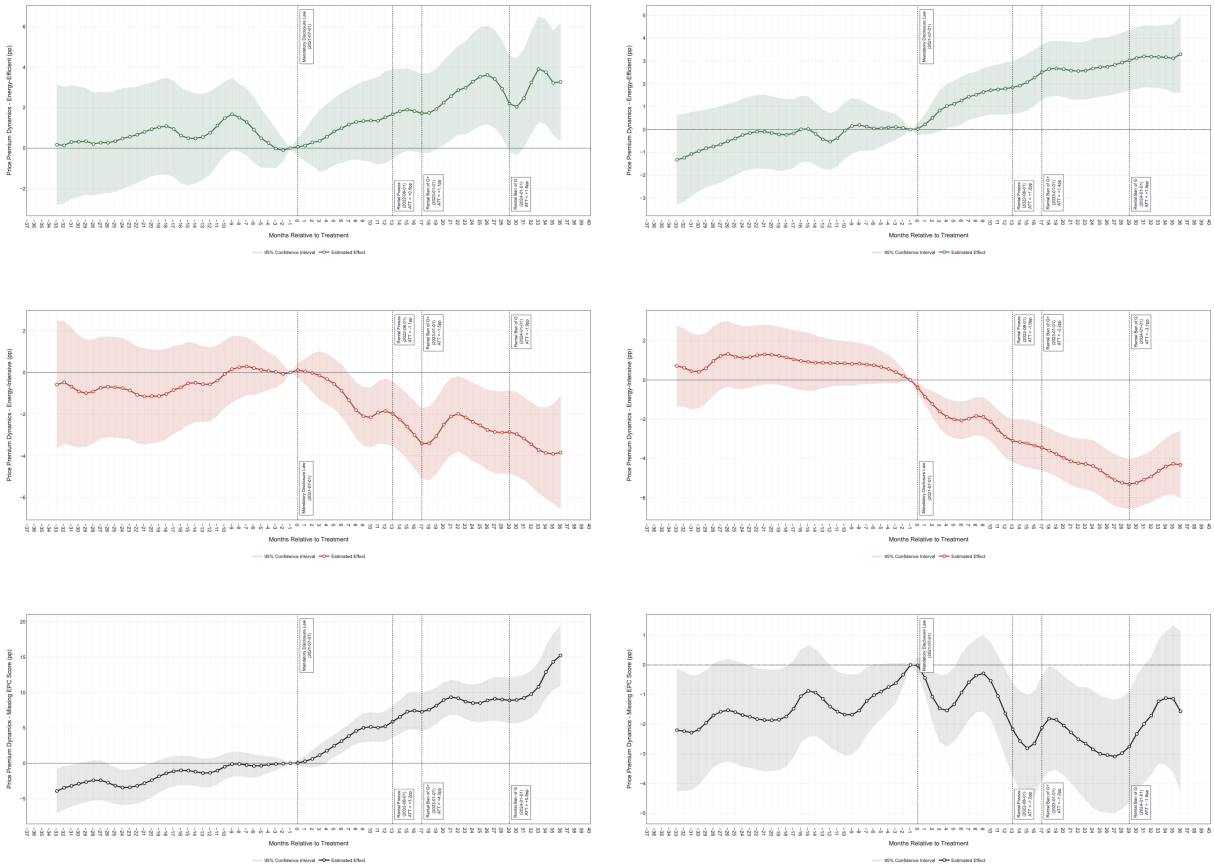
Extending the model to endogenous quality reveals a renovation threshold whose dependence on disclosure enforcement is more subtle than textbook analysis suggests. The renovation decision is naturally framed as a comparison between renovating-and-disclosing and hiding (disclose-low being weakly dominated by hide whenever the non-disclosure pool commands a positive opacity premium). Mechanically, the threshold $\bar{R} = \Pi(d{=}1, \theta{=}1) - \Pi(d{=}0; e^*) + c_0$ appears to scale with the hiding cost. But the equilibrium pool revenue $\Pi(d{=}0; e^*)$ is itself a function of c_0 via the equilibrium hiding rate σ_0^*. In the partial-pooling regime that describes the bulk of French housing markets, the seller's indifference condition $\Pi(d{=}0; e^*) - c_0 = \Pi(d{=}1, \theta{=}0)$ causes the hiding cost to cancel exactly, and the threshold reduces to the verified-quality premium $\Pi(d{=}1, \theta{=}1) - \Pi(d{=}1, \theta{=}0)$, invariant to c_0 and π. The renovation incentive is therefore structurally separate from the non-disclosure premium in the empirically dominant interior regime. This decomposes the policy problem into three instruments rather than the textbook two: renovation subsidies (intensive margin), hiding-cost enforcement (composition of disclosure among the informed), and verification enforcement (the extensive margin of informed sellers, π). The subsidy is the operative renovation instrument in partial pooling. Hiding-cost enforcement does not move the renovation threshold here; the perfect-substitutability of textbook policy analysis holds only in the Complete Adverse Selection corner. Verification enforcement is complementary: it expands the share of the housing stock that is informed and therefore subject to the renovation decision in the first place, and dynamically reclassifies properties from the uninformed pool into the verified strata. The policy lesson is that programmes such as MaPrimeRénov' deliver renovation directly through the verified-quality channel, while audit-production investment is the natural complement that determines how large the renovation base becomes over time. Cross-market targeting should follow ρ (the share of EPC-sensitive buyers), π (audit-infrastructure penetration), and the property-level engineering cost $\hat{R}(X)$. A full welfare ranking, with shadow cost of public funds and an environmental externality, is left to future work; the present analysis establishes the positive incentive object S^* and the regime-conditional substitutability pattern that any welfare exercise must inherit.

A Figures

Figure 11. Dynamic effect of mandatory energy performance disclosure on the listing price of apartments and houses up for sale across various energy consumption levels

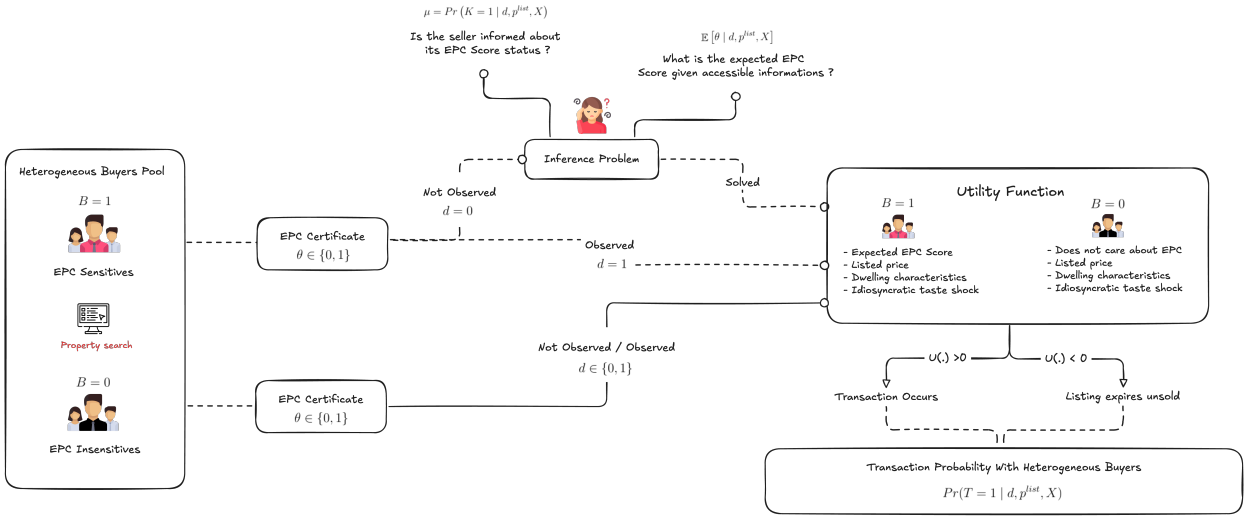
(a) Apartments Up for Sale (Full sample)

(b) Houses Up for Sale (Full sample)



Note. The panels present event-study estimates from the second step of our two-step hedonic procedure (see chapter one). Each series plots the monthly change in hedonic listing prices of the reference property around the introduction of mandatory EPC disclosure, separately by energy-consumption group and property type. Shaded bands represent 95% confidence intervals. These dynamics trace how the policy's effect on predicted apartment and house prices evolves before and after the disclosure requirement in July 2021.

Figure 12. Visual representation of the model dynamics



B Data and Model Objects

Our dataset includes residential property listings from various French housing markets. For each listing i in market m, we observe:

- p_i: listing price
- X_i: vector of property characteristics (such as size, age, location, etc.)
- $d_i \in 0, 1$: indicator of whether the Energy Performance Certificate (EPC) score is displayed in the listing
- $\text{EPC}_i^{\text{listing}}$: the EPC score displayed on the listing (if $d_i = 1$)

What sets this approach apart is our ability to match these listings with data from ADEME (the French Environment and Energy Management Agency), which maintains an official registry of certified energy audits. These audits are reported by third-party assessors who are presumed to be honest and impartial. We match listings to the ADEME registry based on key characteristics that are registered in both databases, such as addresses, floor, item type, and area[8]. This allows us to assess the accuracy of the EPC scores displayed in listings by comparing them to the official certified energy audits. This allows us to observe notably:

- $K \in \{0, 1\}$: whether a verified EPC exists at listing time
- θ_i: the true (verified) EPC score (if $K = 1$)

This matching generates four mutually exclusive groups:

Table 6. Classification of Listings

Group	Displays EPC (d_i)	Verified EPC Exists (K)	Interpretation
Truthful Disclosure	1	1	Informed seller reveals θ
Strategic Hiding	0	1	Informed seller conceals θ
Uninformed	0	0	Seller lacks information
Falsification	1	0 or mismatch	Misreporting (excluded)

[8]We provide a full explanation of our matching strategy in the appendix

The key identification advantage is that we directly observe the latent variable K (information status), which the model treats as unobservable to buyers. This allows us to separately identify the behavior of informed and uninformed sellers.

C Proofs

All results below adopt the benchmark $\alpha_k = \beta_k$.

Preliminary: Optimal Price Sensitivity to Buyer Valuation

Proof C.1 (Optimal price sensitivity to shifts in buyer valuation) *The optimal price p^* satisfies the first-order condition:*

$$\frac{\gamma_1}{\gamma_0 + \gamma_1 \cdot p^*} = 1 - G(v_1 - p^*) \tag{Eq. 25}$$

Define the left-hand side as $\mathrm{LHS}(p) = \frac{\gamma_1}{\gamma_0+\gamma_1 \cdot p}$ and the right-hand side as $\mathrm{RHS}(p; v_1) = 1 - G(v_1 - p)$. The optimal price is determined by the intersection $\mathrm{LHS}(p^*) = \mathrm{RHS}(p^*; v_1)$.

Monotonicity of each side in p.

The left-hand side is strictly decreasing in p:

$$\frac{\partial \mathrm{LHS}}{\partial p} = -\frac{\gamma_1^2}{(\gamma_0 + \gamma_1 \cdot p)^2} < 0 \tag{Eq. 26}$$

The right-hand side is strictly increasing in p. Since G is the logistic CDF, $G'(x) = G(x)(1 - G(x)) > 0$ for all x. Therefore:

$$\frac{\partial \mathrm{RHS}}{\partial p} = G'(v_1 - p) > 0 \tag{Eq. 27}$$

A strictly decreasing function intersects a strictly increasing function at most once, so p^* is unique.

Effect of v_1 on the right-hand side.

At any fixed p, the right-hand side is strictly decreasing in v_1:

$$\frac{\partial \mathrm{RHS}}{\partial v_1} = -G'(v_1 - p) < 0 \tag{Eq. 28}$$

An increase in v_1 shifts the RHS curve downward.

Comparative static.

Using the chain rule, differentiating the equilibrium condition with respect to v_1:

$$\frac{\partial \mathrm{LHS}}{\partial p} \cdot \frac{\partial p^*}{\partial v_1} = \frac{\partial \mathrm{RHS}}{\partial p} \cdot \frac{\partial p^*}{\partial v_1} + \frac{\partial \mathrm{RHS}}{\partial v_1} \tag{Eq. 29}$$

Rearranging:

$$\frac{\partial p^*}{\partial v_1} = \frac{\frac{\partial \mathrm{RHS}}{\partial v_1}}{\frac{\partial \mathrm{LHS}}{\partial p} - \frac{\partial \mathrm{RHS}}{\partial p}} = \frac{-G'(v_1 - p^*)}{-\frac{\gamma_1^2}{(\gamma_0+\gamma_1 \cdot p^*)^2} - G'(v_1 - p^*)} > 0 \tag{Eq. 30}$$

The numerator is negative. The denominator is negative (sum of two negative terms). Therefore $\frac{\partial p^*}{\partial v_1} > 0$.

Lemma A.1 (Profit is Increasing in Sensitive-Buyer Valuation)
Holding v_0 and ρ fixed, the gross profit function $\tilde{\Pi}(v_1; v_0, \rho)$ is strictly increasing in v_1 whenever $\rho > 0$.

Proof C.2 (Proof of Lemma A.1) *Let $p^*_{\text{list}}(v_1)$ denote the optimal price given sensitive-buyer valuation v_1. By the envelope theorem, the effect of a small increase in v_1 on maximized profit is:*

$$\frac{\partial \tilde{\Pi}}{\partial v_1} = \phi(p^*_{\text{list}}) \cdot \rho \cdot G'(v_1 - p^*_{\text{list}}) \tag{Eq. 31}$$

where we evaluate the partial derivative of the objective holding p fixed at the optimum.

All three terms are positive: $\phi(p^*) > 0$ (expected transaction price is positive), $\rho \in [0, 1] \geq 0$ (by assumption), and $G'(\cdot) > 0$ (logistic density is everywhere positive). Therefore $\frac{\partial \tilde{\Pi}}{\partial v_1} > 0$.

Since $\tilde{\Pi}$ is increasing in v_1, the profit gain from disclosure is positive if and only if $v_1(d = 1) > v_1(d = 0)$: sensitive buyers value the property more when quality is revealed than when it is hidden. In this case, disclosure raises profit, and the seller discloses (absent hiding costs). Conversely, the profit gain is negative if $v_1(d = 1) < v_1(d = 0)$: sensitive buyers value the property more under hiding. This occurs when buyers' inference about non-disclosers is more favorable than the truth. In this case, hiding raises profit, and the seller hides (absent sufficiently high hiding costs).

Proof of Proposition 4.1 (Disclosure Incentives and Seller Sorting)

Proof C.3 *By Lemma A.1, $p^*_0 > p^*_1$ if and only if $v_1(d = 0) > v_1(d = 1, \theta)$. Since $v_1(d = 0) - v_1(d = 1, \theta) = \beta_1 \cdot (\mathbb{E}[\theta \mid d = 0] - \theta)$ and $\beta_1 > 0$, this holds if and only if $\mathbb{E}[\theta \mid d = 0] > \theta$.*

- **For a low-quality seller** *($\theta = 0$): the condition is $\mathbb{E}[\theta \mid d = 0] > 0$, which holds whenever the non-disclosing pool contains any high-quality sellers.*
- **For a high-quality seller** *($\theta = 1$): the condition is $\mathbb{E}[\theta \mid d = 0] > 1$, which is impossible since $\theta \in \{0, 1\}$.*

Proof of Corollary 4.1 (High-Quality Sellers Always Disclose)

Proof C.4 *From Proposition 4.1, the non-disclosure premium exists if and only if $\mathbb{E}[\theta \mid d = 0] > \theta$.*

For a high-quality seller ($\theta = 1$), this condition requires $\mathbb{E}[\theta \mid d = 0] > 1$. But $\theta \in \{0, 1\}$, so $\mathbb{E}[\theta \mid d = 0] \leq 1$ always. Therefore the condition never holds: high-quality sellers never earn a premium from hiding.

In fact, the inequality is strict whenever the non-disclosing pool contains any low-quality sellers. In that case $\mathbb{E}[\theta \mid d = 0] < 1$, so:

$$v_1(d = 1, \theta = 1) - v_1(d = 0) = \beta_1 \cdot (1 - \mathbb{E}[\theta \mid d = 0]) > 0 \tag{Eq. 32}$$

Disclosure yields a strictly higher valuation, and therefore strictly higher profit. Adding hiding costs $C(\theta = 1) = c_0 + c_1 \geq 0$ only reinforces this: disclosure dominates hiding.

Proof of Proposition 4.2 (Equilibrium Existence and Uniqueness)

Proof C.5 *By Corollary 4.1, $\sigma^*_1 = 0$ in any equilibrium. The strategic problem reduces to characterising $\sigma^*_0 \in [0, 1]$, the hiding probability of informed low-quality sellers. The incentive to hide is*

$$\Phi(\sigma_0) \equiv \Delta(e(\sigma_0)) - c_0 = \frac{\rho\, \beta_1\, (1 - \pi)\bar{\theta}}{\pi(1 - \bar{\theta})\sigma_0 + (1 - \pi)} - c_0,$$

where $e(\sigma_0)$ is the equilibrium pool belief (Equation (Eq. 5)). Since $e(\sigma_0)$ is strictly decreasing in σ_0 (Section 6.3) and Δ is strictly increasing in e, Φ is strictly decreasing *in σ_0, with boundary values*

$$\Phi(0) = \Delta(\bar{\theta}) - c_0, \qquad \Phi(1) = \Delta\left(\tfrac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}}\right) - c_0.$$

Case 1: Cost-Deterred Hiding. *Suppose $c_0 > \Delta(\bar{\theta})$, i.e., $\Phi(0) < 0$.*

Since Φ is strictly decreasing, $\Phi(\sigma_0) \leq \Phi(0) < 0$ for all $\sigma_0 \in [0,1]$. Hiding is strictly unprofitable at every hiding rate, including the most favourable case $\sigma_0 = 0$. Disclosure strictly dominates for every informed low-quality seller, so the unique equilibrium is $\sigma_0^* = 0$ with equilibrium belief $e^* = \bar{\theta}$.

Interpretation. All informed low-quality sellers disclose; the non-disclosing pool consists exclusively of uninformed sellers. Buyers correctly infer quality $\bar{\theta}$ from non-disclosure. Unlike the classical Grossman–Milgrom outcome, silence is uninformative rather than damning: non-disclosure carries no stigma because it reveals nothing beyond the seller's lack of a certified audit.

Case 2: Complete Adverse Selection. *Suppose $c_0 < \Delta(e(1))$, i.e., $\Phi(1) > 0$.*

Since Φ is strictly decreasing, $\Phi(\sigma_0) \geq \Phi(1) > 0$ for all $\sigma_0 \in [0,1]$. Hiding is strictly profitable at every hiding rate, including the least favourable case $\sigma_0 = 1$. Hiding is therefore strictly dominant for every informed low-quality seller regardless of the strategies of others, so the unique equilibrium is $\sigma_0^* = 1$ with equilibrium belief $e^* = (1-\pi)\bar{\theta}/(1-\pi\bar{\theta})$.

Interpretation. All informed low-quality sellers pool with uninformed sellers. The non-disclosure premium $\rho \cdot \beta_1 \cdot e(1)$ is the maximum attainable across all regimes (Equation (Eq. 8)).

Case 3: Partial Pooling. *Suppose $\Delta(e(1)) \leq c_0 \leq \Delta(\bar{\theta})$, i.e., $\Phi(1) \leq 0 \leq \Phi(0)$.*

Φ is continuous and strictly decreasing on $[0,1]$, with $\Phi(0) \geq 0$ and $\Phi(1) \leq 0$. By the intermediate value theorem, there exists $\sigma_0^* \in [0,1]$ with $\Phi(\sigma_0^*) = 0$; strict monotonicity of Φ implies uniqueness. When the boundary inequalities hold strictly, σ_0^* lies in the open interval $(0,1)$; at the boundaries, $\sigma_0^* \in \{0,1\}$ and the regime collapses to Case 1 or Case 2.

At σ_0^*, informed low-quality sellers are exactly indifferent between hiding and disclosing; mixing at rate σ_0^* is self-confirming through the belief function. This regime generates the closed-form hiding rate and is the empirically dominant regime in the estimation sample.

Uniqueness and exhaustiveness. *In Cases 1 and 2, uniqueness follows from strict dominance: Φ has constant sign on $[0,1]$ so no other σ_0 is consistent with optimality. In Case 3, uniqueness follows from the unique zero of Φ. The three conditions $c_0 > \Delta(\bar{\theta})$, $c_0 < \Delta(e(1))$, and $\Delta(e(1)) \leq c_0 \leq \Delta(\bar{\theta})$ partition $\mathbb{R}_+$ since $\Delta(\bar{\theta}) \geq \Delta(e(1))$ (as $e(0) = \bar{\theta} \geq e(1)$ and Δ is increasing). Equilibrium exists and is unique for all parameter values.*

Proof of Non-Disclosure Premium Characterisation (Eq. 8)

Proof C.6 *Under the benchmark $\alpha_k = \beta_k$, the equilibrium price of a listing equals its expected buyer valuation. For a low-quality seller who discloses ($d = 1$, $\theta = 0$), both buyer types value the property at β_0 (sensitive buyers: $v_1(d=1, \theta=0) = \beta_0 + \beta_1 \cdot 0 = \beta_0$; insensitive buyers: $v_0 = \beta_0$), so $p_1^* = \beta_0$. For a non-disclosing listing, sensitive buyers assign expected quality $e(\sigma_0^*)$ and value the property at $\beta_0 + \beta_1 e(\sigma_0^*)$, while insensitive buyers value it at β_0. Weighting by buyer composition yields $p_0^* = \beta_0 + \rho\,\beta_1\,e(\sigma_0^*)$, so the non-disclosure premium is:*

$$p_0^* - p_1^* = \rho\,\beta_1\,e(\sigma_0^*). \qquad \text{(Eq. 33)}$$

The three cases of (Eq. 8) follow by substituting the equilibrium σ_0^* from each regime of Proposition 4.2 into (Eq. 33).

Case 1: Cost-Deterred Hiding ($\sigma_0^* = 0$). *Evaluating (Eq. 5) at $\sigma_0 = 0$:*

$$e(0) = \frac{(1-\pi)\bar{\theta}}{(1-\pi)} = \bar{\theta}.$$

Substituting into (Eq. 33): $p_0^* - p_1^* = \rho\,\beta_1\,\bar{\theta}$.

Case 2: Complete Adverse Selection ($\sigma_0^* = 1$). *With $\sigma_1^* = 0$ by Corollary 4.1, evaluating (Eq. 5) at $\sigma_0 = 1$:*

$$e(1) = \frac{(1-\pi)\bar{\theta}}{\pi(1-\bar{\theta}) + (1-\pi)} = \frac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}}.$$

Substituting into (Eq. 33)*:* $p_0^* - p_1^* = \rho\,\beta_1\,\dfrac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}}$.

Case 3: Partial Pooling ($\sigma_0^* \in (0,1)$). *By Proposition 4.2, σ_0^* solves the indifference condition $\Delta(e(\sigma_0^*)) = c_0$, i.e., $\rho\,\beta_1\,e(\sigma_0^*) = c_0$. But the left-hand side is $p_0^* - p_1^*$ by* (Eq. 33)*, so $p_0^* - p_1^* = c_0$.*

Ordering. *Since e is strictly decreasing by Lemma 3.1, we have $e(1) < e(\sigma_0^*) < e(0)$ for all $\sigma_0^* \in (0,1)$, hence $\rho\,\beta_1\,e(1) < c_0 < \rho\,\beta_1\,\bar{\theta}$, confirming the strict ordering stated after* (Eq. 8)*.*

Proof of Proposition 4.3 (Comparative statics of the non-disclosure premium)

Proof C.7 *From* (Eq. 33)*, the non-disclosure premium is $\Delta \equiv p_0^* - p_1^* = \rho\,\beta_1\,e(\sigma_0^*)$, where the belief $e(\sigma_0)$ is given by* (Eq. 5)*. The equilibrium hiding rate $\sigma_0^* \in [0,1]$ is determined by the regime characterised in Proposition 4.2. We derive $\partial\Delta/\partial\pi$ and $\partial\Delta/\partial\rho$ in each of the three regimes, in turn.*

Case 1: Cost-Deterred Hiding ($\sigma_0^* = 0$). *No informed low-quality seller hides, so the non-disclosing pool consists entirely of uninformed sellers. From* (Eq. 5) *at $\sigma_0 = 0$:*

$$e(0) = \frac{(1-\pi)\bar{\theta}}{(1-\pi)} = \bar{\theta}, \tag{Eq. 34}$$

which is independent of π. The premium is therefore:

$$\Delta = \rho\,\beta_1\,\bar{\theta}. \tag{Eq. 35}$$

Since $\bar{\theta}$ is a structural parameter, $\partial\Delta/\partial\pi = 0$ and $\partial\Delta/\partial\rho = \beta_1\bar{\theta} > 0$.

Case 2: Complete Adverse Selection ($\sigma_0^* = 1$). *All informed low-quality sellers hide. From* (Eq. 5) *at $\sigma_0 = 1$:*

$$e(1) = \frac{(1-\pi)\bar{\theta}}{\pi(1-\bar{\theta}) + (1-\pi)} = \frac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}}, \tag{Eq. 36}$$

so $\Delta = \rho\,\beta_1\,e(1)$. Differentiating with respect to π by the quotient rule:

$$\frac{\partial e(1)}{\partial\pi} = \frac{-\bar{\theta}(1-\pi\bar{\theta}) - (1-\pi)\bar{\theta}\cdot(-\bar{\theta})}{(1-\pi\bar{\theta})^2} = \frac{\bar{\theta}(\bar{\theta}-1)}{(1-\pi\bar{\theta})^2} = \frac{-\bar{\theta}(1-\bar{\theta})}{(1-\pi\bar{\theta})^2} < 0, \tag{Eq. 37}$$

hence $\partial\Delta/\partial\pi = \rho\,\beta_1 \cdot \partial e(1)/\partial\pi < 0$. Differentiating with respect to ρ (with $e(1)$ determined solely by π and $\bar{\theta}$): $\partial\Delta/\partial\rho = \beta_1\,e(1) > 0$.

Case 3: Partial Pooling ($\sigma_0^* \in (0,1)$). *The interior equilibrium satisfies $c_0 = \rho\,\beta_1\,e(\sigma_0^*)$, so the premium is pinned at the hiding cost:*

$$\Delta = c_0. \tag{Eq. 38}$$

Changes in π or ρ induce compensating adjustments in σ_0^* that restore the indifference condition, leaving the premium invariant: $\partial\Delta/\partial\pi = 0$ and $\partial\Delta/\partial\rho = 0$.

To verify: from (Eq. 5)*, $\partial e(\sigma_0)/\partial\pi = -\bar{\theta}(1-\bar{\theta})\sigma_0/[\pi(1-\bar{\theta})\sigma_0 + (1-\pi)]^2 < 0$ for $\sigma_0 > 0$. Total differentiation of $c_0 = \rho\,\beta_1\,e(\sigma_0^*(\pi),\pi)$ with respect to π gives:*

$$0 = \rho\,\beta_1\left(\frac{\partial e}{\partial\sigma_0}\,\frac{\mathrm{d}\sigma_0^*}{\mathrm{d}\pi} + \frac{\partial e}{\partial\pi}\right), \tag{Eq. 39}$$

so $\mathrm{d}\sigma_0^*/\mathrm{d}\pi = -(\partial e/\partial\pi)/(\partial e/\partial\sigma_0) < 0$: as π rises, informed sellers populate the non-disclosing pool more heavily, so fewer low-quality sellers need to hide to sustain the same buyer belief, and σ_0^* falls.

Summary. *The non-disclosure premium is* weakly *decreasing in π and* weakly *increasing in ρ, with strict monotonicity only in the Complete Adverse Selection regime. In Case 1, the premium is determined by the population quality $\bar{\theta}$ of the uninformed pool and is insensitive to market fundamentals π and ρ. In Case 3, the premium is pinned at the primitive c_0 by the indifference condition; it is the hiding rate σ_0^*, not the premium, that absorbs variation in π and ρ.*

Proof of Corollary ?? (Effect of ρ on the non-disclosure premium)

Proof C.8 Part (i). *From* (Eq. 33)*, the non-disclosure premium is* $\Delta = \rho\,\beta_1\,e(\sigma_0^*)$*. We differentiate with respect to* ρ *in each regime, holding the regime-specific equilibrium* σ_0^* *fixed where appropriate.*

Cost-Deterred Hiding ($\sigma_0^* = 0$*). Here* $e(0) = \bar{\theta}$ *by* (Eq. 5)*, independent of* ρ*. Therefore* $\partial\Delta/\partial\rho = \beta_1\bar{\theta} > 0$*.*

Complete Adverse Selection ($\sigma_0^* = 1$*). Here* $e(1) = (1-\pi)\bar{\theta}/(1-\pi\bar{\theta})$*, also independent of* ρ*. Therefore* $\partial\Delta/\partial\rho = \beta_1\,e(1) > 0$*.*

Partial Pooling ($\sigma_0^* \in (0,1)$*). The indifference condition* $\rho\,\beta_1\,e(\sigma_0^*) = c_0$ *pins* $\Delta = c_0$*. Changes in* ρ *induce compensating adjustments in* σ_0^**, as established in Proposition 4.3(ii):* $\partial\sigma_0^*/\partial\rho > 0$*. The premium satisfies* $\partial\Delta/\partial\rho = 0$*.*

Part (ii). *From Proposition 4.2, the* Partial Pooling *regime requires* $\Delta(e(1)) \le c_0 \le \Delta(\bar{\theta})$*, i.e.,* $\rho\,\beta_1\,e(1) \le c_0 \le \rho\,\beta_1\,\bar{\theta}$*. Differentiating both bounds with respect to* ρ*:* $\partial[\rho\,\beta_1\,e(1)]/\partial\rho = \beta_1\,e(1) > 0$ *and* $\partial[\rho\,\beta_1\,\bar{\theta}]/\partial\rho = \beta_1\,\bar{\theta} > 0$*. Both boundaries increase with* ρ*. The width of the partial-pooling interval is* $\rho\,\beta_1\,[\bar{\theta} - e(1)]$*, which is increasing in* ρ *since* $\bar{\theta} > e(1)$ *whenever* $\pi > 0$*. The regime is therefore weakly expanding in* ρ*.*

Proof of Proposition ?? (Equilibrium effect of π: equilibrium moves toward full unraveling)

Proof C.9 *From Eq.* (Eq. 5)*, the equilibrium belief under partial separation is:*

$$e(1) = \frac{(1-\pi)\bar{\theta}}{1-\pi\bar{\theta}} \tag{Eq. 40}$$

Taking the derivative:

$$\frac{\partial e(1)}{\partial\pi} = \frac{-\bar{\theta}(1-\pi\bar{\theta}) - (1-\pi)\bar{\theta}(-\bar{\theta})}{(1-\pi\bar{\theta})^2} = \frac{-\bar{\theta}(1-\bar{\theta})}{(1-\pi\bar{\theta})^2} < 0 \tag{Eq. 41}$$

As π *increases,* $e(1)$ *falls: the pool quality under partial separation deteriorates because fewer uninformed sellers are available to provide cover.*

Since $\Delta(e)$ *is increasing in* e*,* $\Delta(e(1))$ *falls. The condition for partial separation,* $c_0 < \Delta(e(1))$*, becomes harder to satisfy. For fixed* c_0*, an increase in* π *can push the equilibrium from partial separation into the mixed or full unraveling region.*

When most sellers are informed (π high), non-disclosure is suspicious. Buyers recognize that uninformed sellers are rare, so most non-disclosers must be strategic hiders. The favorable cover provided by uninformed sellers evaporates. Low-quality sellers find hiding less attractive, and the market approaches classical unraveling.

When most buyers care about quality (ρ high), the stakes of disclosure are high. Low-quality sellers have much to lose from revealing their type. The non-disclosure premium is large, making hiding attractive even when pool quality is low. Partial separation is sustained, and by Corollary **??**(ii), the partial-pooling interval itself widens with ρ.

Proof of Proposition ?? (Equilibrium effect of c_0: equilibrium moves toward full unraveling)

Proof C.10 *An increase in* c_0 *shifts the function* $\Phi(\sigma_0) = \Delta(e(\sigma_0)) - c_0$ *downward. The boundaries become:*

$$\begin{aligned} \Phi(0) &= \Delta(\bar{\theta}) - c_0 \quad \textit{(decreases)} \\ \Phi(1) &= \Delta(e(1)) - c_0 \quad \textit{(decreases)} \end{aligned} \tag{Eq. 42}$$

The condition for full unraveling, $c_0 > \Delta(\bar{\theta})$*, becomes easier to satisfy. The condition for partial separation,* $c_0 < \Delta(e(1))$*, becomes harder to satisfy.*

Higher hiding costs make concealment less attractive. Regulatory enforcement, administrative burden, or reputational concerns that raise c_0 push the market toward disclosure.

Proof of Lemma 8.1 (Equilibrium reduction in partial pooling)

Proof C.11 *Write $\Pi(d=0) \equiv \Pi(d=0;e^*)$ where the equilibrium belief is $e^* = e(\sigma_0^*)$ from (Eq. 5). The revenue $\Pi(d=1,\theta)$ is independent of σ_0^* since buyers observe θ directly under disclosure (Corollary 4.1).*

Partial Pooling. By Proposition 4.2, in the interior regime $\sigma_0^* \in (0,1)$ is determined by the indifference condition

$$\Pi(d=0;e^*) - c_0 \;=\; \Pi(d=1,\theta=0), \tag{Eq. 43}$$

which states that an informed low-quality seller is indifferent between hiding (paying c_0 for the inflated pool revenue) and disclosing low quality. Substituting (Eq. 43) into the threshold definition (Eq. 21),

$$\bar{R}^{PP} \;=\; \Pi(d=1,\theta=1) - \Pi(d=0;e^*) + c_0 \;=\; \Pi(d=1,\theta=1) - \Pi(d=1,\theta=0).$$

The cancellation is exact: the apparent $+c_0$ in the threshold formula is offset by the equilibrium movement of $\Pi(d=0;e^*)$ (which in PP exceeds $\Pi(d=1,\theta=0)$ by exactly c_0). $\bar{R}^{PP}$ depends on ρ and X only.

Complete Adverse Selection. The regime $\sigma_0^* = 1$ obtains because hiding strictly dominates disclose-low at the equilibrium pool composition $e^{CAS} = (1-\pi)\bar{\theta}/(1-\pi\bar{\theta})$: $\Pi(d=0;e^{CAS}) - c_0 > \Pi(d=1,\theta=0)$. The corner is locally stable in c_0 within the regime's defining inequality, so σ_0^* and e^* are locally constant in c_0, and the threshold retains its primitive form $\bar{R}^{CAS} = \Pi(d=1,\theta=1) - \Pi(d=0;e^{CAS}) + c_0$ with $\partial\bar{R}^{CAS}/\partial c_0 = 1$.

Proof of Proposition 8.1 (Subsidy comparative statics by regime)

Throughout, we use the regime-conditional form of $\bar{R}$ from Lemma 8.1. We assume G is log-concave so that $\Pi(d,\theta;e) = \max_p \phi(p)\Gamma(p \mid d,\theta;e)$ admits a unique interior maximiser $p^*(d,\theta;e)$ and is differentiable in e. The optimal subsidy $S^* = \max\{\hat{R}(X) - \bar{R}, 0\}$ is piecewise linear in its arguments; we evaluate the derivatives on the active region $\hat{R}(X) > \bar{R}$. The relation $\partial S^*/\partial\bar{R} = -1$ then translates each derivative of $\bar{R}$ into the corresponding derivative of S^*.

Partial Pooling

Proof C.12 (Effect of c_0) *From Lemma 8.1, in PP, $\bar{R}^{PP} = \Pi(d=1,\theta=1) - \Pi(d=1,\theta=0)$. Both terms are verified-disclosure revenues with θ observable; neither depends on the equilibrium hiding rate, the non-disclosing pool, or the cost of hiding. Hence $\partial\bar{R}^{PP}/\partial c_0 = 0$, and consequently $\partial S^*/\partial c_0 = 0$ when $\hat{R}(X) > \bar{R}^{PP}$.*

Mechanism. The equilibrium indifference (Eq. 43) forces $\Pi(d=0;e^*)$ to track c_0 one-for-one through the equilibrium response of σ_0^*, so the apparent $+c_0$ in the primitive threshold formula (Eq. 21) is exactly cancelled by the equilibrium movement of $\Pi(d=0;e^*)$. The renovation incentive reduces to the verified-quality premium, which involves no hiding-related object.

Proof C.13 (Effect of π) *The verified-quality revenues $\Pi(d=1,\theta\in\{0,1\})$ are independent of the non-disclosing pool composition and hence of π. Therefore $\partial\bar{R}^{PP}/\partial\pi = 0$ and $\partial S^*/\partial\pi = 0$.*

Proof C.14 (Effect of ρ) *Buyer valuations under disclosure satisfy $v_1(d=1,\theta) = \beta_0 + \beta_1\theta + \sum_n \beta_n X_n$ and $v_0 = \beta_0 + \sum_n \beta_n X_n$. The aggregate demand under disclosure is*

$$\Gamma(p \mid d=1,\theta) = \rho\, G(v_1(d=1,\theta) - p) + (1-\rho)\, G(v_0 - p).$$

By the envelope theorem applied to $\Pi(d=1,\theta) = \max_p \phi(p)\Gamma(p \mid d=1,\theta)$,

$$\frac{\partial\Pi(d=1,\theta)}{\partial\rho} = \phi(p^*)\left[G(v_1(d=1,\theta) - p^*) - G(v_0 - p^*)\right].$$

Subtracting the $\theta=0$ from the $\theta=1$ case (noting $v_1(d=1,\theta=0) = v_0$, so the bracket is zero at $\theta=0$ and strictly positive at $\theta=1$ since $\beta_1 > 0$ and G is strictly increasing):

$$\frac{\partial\bar{R}}{\partial\rho} = \frac{\partial\Pi(d=1,\theta=1)}{\partial\rho} - \frac{\partial\Pi(d=1,\theta=0)}{\partial\rho} > 0,$$

where the optimal price p^* in each term is evaluated at its own argument. Consequently $\partial S^*/\partial\rho \le 0$ in the active region.

Complete Adverse Selection

Proof C.15 (Effect of c_0) *In CAS, $\sigma_0^* = 1$ is locally constant in c_0 within the regime's defining inequality, hence e^{CAS} is independent of c_0 and so is $\Pi(d=0; e^{CAS})$. Differentiating $\bar{R}^{CAS} = \Pi(d=1, \theta=1) - \Pi(d=0; e^{CAS}) + c_0$ directly yields $\partial \bar{R}/\partial c_0 = 1$, hence $\partial S^*/\partial c_0 = -1$ when $\hat{R}(X) > \bar{R}$. The textbook substitutability between hiding-cost enforcement and renovation subsidies thus holds only at this corner.*

Proof C.16 (Effect of π) *In CAS, $e^{CAS} = (1-\pi)\bar{\theta}/(1-\pi\bar{\theta})$ with $\partial e^{CAS}/\partial \pi < 0$. By the envelope theorem,*

$$\frac{\partial \Pi(d=0; e^{CAS})}{\partial \pi} = \frac{\partial \Pi(d=0; e^{CAS})}{\partial e} \cdot \frac{\partial e^{CAS}}{\partial \pi},$$

with $\partial \Pi/\partial e = \phi(p^*)\rho\beta_1\, g(v_1(d=0) - p^*) > 0$ (because $v_1(d=0) = \beta_0 + \beta_1 e + \sum_n \beta_n X_n$ is increasing in e and G is strictly increasing). Hence $\partial \Pi(d=0; e^{CAS})/\partial \pi < 0$ and $\partial \bar{R}/\partial \pi = -\partial \Pi(d=0; e^{CAS})/\partial \pi > 0$, giving $\partial S^*/\partial \pi \leq 0$.

Proof C.17 (Effect of ρ) *The argument follows the PP case, with the additional step that $\Pi(d=0; e^{CAS})$ depends on ρ through the share of EPC-sensitive buyers in aggregate demand; e^{CAS} itself is independent of ρ. Applying the envelope theorem to both $\Pi(d=1, \theta=1)$ and $\Pi(d=0; e^{CAS})$ and using $v_1(d=1, \theta=1) > v_1(d=0)$ yields $\partial \bar{R}^{CAS}/\partial \rho > 0$.*

Effect of renovation cost $\hat{R}(X)$

Proof C.18 *In the active region $\hat{R}(X) > \bar{R}$, the unconstrained form $S^* = \hat{R}(X) - \bar{R}$ applies, giving $\partial S^*/\partial \hat{R}(X) = 1$. In the inactive region, $S^* = 0$ and the derivative is zero. The indicator-function statement in Proposition 8.1 captures both.*

Summary of policy reading. The substitutability $\partial S^*/\partial c_0 = -1$ obtains only in the Complete Adverse Selection corner, where the equilibrium hiding rate is at the boundary and $\Pi(d=0; e^*)$ is locally constant in c_0. In the partial-pooling regime, which describes the bulk of French apartment and house markets, the equilibrium indifference identity forces $\Pi(d=0; e^*)$ to move one-for-one with c_0, so the apparent $+c_0$ in the threshold formula is cancelled exactly and $\partial S^*/\partial c_0 = 0$. Disclosure enforcement and renovation subsidies are therefore policy complements rather than substitutes outside the CAS corner: enforcement shifts the within-stock composition of hiders and disclosers along the indifference curve, while subsidies shift the stock itself toward verified high quality.

References

Hunt Allcott and Michael Greenstone. Is There an Energy Efficiency Gap? *Journal of Economic Perspectives*, 26(1):3–28, February 2012. ISSN 0895-3309. doi: 10.1257/jep.26.1.3. URL https://www.aeaweb.org/articles?id=10.1257/jep.26.1.3.

Omri Ben-Shahar and Carl E. Schneider. The Failure of Mandated Disclosure. *University of Pennsylvania Law Review*, 159(3):647–749, 2011. ISSN 0041-9907. URL https://scholarship.law.upenn.edu/penn_law_review/vol159/iss3/1/.

Dirk Brounen and Nils Kok. On the Economics of Energy Labels in the Housing Market. *Journal of Environmental Economics and Management*, 62(2):166–179, 2011. ISSN 0095-0696. doi: 10.1016/j.jeem.2011.05.006. URL https://www.sciencedirect.com/science/article/pii/S0095069611000691.

Ronald A. Dye. Disclosure of Nonproprietary Information. *Journal of Accounting Research*, 23(1):123–145, 1985. ISSN 0021-8456. doi: 10.2307/2490910. URL https://www.jstor.org/stable/2490910.

Michael J. Fishman and Kathleen M. Hagerty. Mandatory versus Voluntary Disclosure in Markets with Informed and Uninformed Customers. *Journal of Law, Economics, & Organization*, 19(1):45–63, 2003. ISSN 8756-6222. doi: 10.1093/jleo/19.1.45. URL https://www.jstor.org/stable/3555046.

Manuel Frondel, Andreas Gerster, and Colin Vance. The Power of Mandatory Quality Disclosure: Evidence from the German Housing Market, May 2018. URL https://papers.ssrn.com/abstract=2796532.

Franz Fuerst, Pat McAllister, Anupam Nanda, and Peter Wyatt. Energy Performance Ratings and House Prices in Wales: An Empirical Study. *Energy Policy*, 92:20–33, 2016. ISSN 0301-4215. doi: 10.1016/j.enpol.2016.01.024. URL https://www.sciencedirect.com/science/article/pii/S0301421516300970.

Todd D. Gerarden, Richard G. Newell, and Robert N. Stavins. Assessing the Energy-Efficiency Gap. *Journal of Economic Literature*, 55(4):1486–1525, December 2017. ISSN 0022-0515. doi: 10.1257/jel.20161360. URL https://www.aeaweb.org/articles?id=10.1257/jel.20161360.

Kenneth Gillingham and Karen Palmer. Bridging the Energy Efficiency Gap: Policy Insights from Economic Theory and Empirical Evidence. *Review of Environmental Economics and Policy*, 8(1):18–38, 2014. ISSN 1750-6816. doi: 10.1093/reep/ret021. URL https://academic.oup.com/reep/article/8/1/18/1558137.

Sanford J. Grossman. The Informational Role of Warranties and Private Disclosure about Product Quality. *The Journal of Law and Economics*, 24(3):461–483, December 1981. ISSN 0022-2186. doi: 10.1086/466995. URL https://www.journals.uchicago.edu/doi/abs/10.1086/466995. Publisher: The University of Chicago Press.

Marie Hyland, Ronan C. Lyons, and Seán Lyons. The Value of Domestic Building Energy Efficiency – Evidence from Ireland. *Energy Economics*, 40:943–952, 2013. ISSN 0140-9883. doi: 10.1016/j.eneco.2013.06.020. URL https://www.sciencedirect.com/science/article/pii/S0140988313001096.

Woojin Jung and Young K. Kwon. Disclosure When the Market is Unsure of Information Endowment of Managers. *Journal of Accounting Research*, 26(1):146–153, 1988. ISSN 0021-8456. doi: 10.2307/2491118. URL https://www.jstor.org/stable/2491118.

Steven D. Levitt and Chad Syverson. Market Distortions when Agents are Better Informed: The Value of Information in Real Estate Transactions. *Review of Economics and Statistics*, 90(4):599–611, 2008. ISSN 0034-6535. doi: 10.1162/rest.90.4.599. URL https://www.mitpressjournals.org/doi/abs/10.1162/rest.90.4.599.

Alessandro Lizzeri. Information Revelation and Certification Intermediaries. *RAND Journal of Economics*, 30(2):214–231, 1999. ISSN 0741-6261. doi: 10.2307/2556067. URL https://www.jstor.org/stable/2556067.

George Loewenstein, Cass R. Sunstein, and Russell Golman. Disclosure: Psychology Changes Everything. *Annual Review of Economics*, 6(1):391–419, August 2014. ISSN 1941-1383, 1941-1391. doi: 10.1146/annurev-economics-080213-041341. URL https://www.annualreviews.org/doi/10.1146/annurev-economics-080213-041341.

Paul Milgrom. Good News and Bad News: Representation Theorems and Applications. *Bell Journal of Economics*, 12(2):380–391, 1981. ISSN 0361-915X. URL https://econpapers.repec.org/article/rjebellje/v_3a12_3ay_3a1981_3ai_3aautumn_3ap_3a380-391.htm. Publisher: The RAND Corporation.

Erica Myers. Asymmetric information in residential rental markets: Implications for the energy efficiency gap. *Journal of Public Economics*, 190:104251, October 2020. ISSN 0047-2727. doi: 10.1016/j.jpubeco.2020.104251. URL https://www.sciencedirect.com/science/article/pii/S0047272720301158.

Erica Myers, Steven L. Puller, and Jeremy West. Mandatory Energy Efficiency Disclosure in Housing Markets. *American Economic Journal: Economic Policy*, 14(4):453–487, November 2022. ISSN 1945-7731, 1945-774X. doi: 10.1257/pol.20200539. URL https://pubs.aeaweb.org/doi/10.1257/pol.20200539.

Marie-Laure Nauleau. Free-Riding on Tax Credits for Home Insulation in France: An Econometric Assessment Using Panel Data. *Energy Economics*, 46:78–92, 2014. ISSN 0140-9883. doi: 10.1016/j.eneco.2014.08.007. URL https://www.sciencedirect.com/science/article/pii/S0140988314001571.